WEIL'S QUADRATIC FORM VIA THE SCREW FUNCTION

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ABSTRACT. We establish a unified operator-theoretic framework for understanding the results on the Weil quadratic form obtained by Yoshida (1992), Bombieri (2001, 2003), Connes–Consani (2023), and Connes–Consani–Moscovici (2025+) from the perspective of the screw function introduced in Suzuki (2023). An advantage of the approach via the screw function is that it provides a method to study the Weil quadratic form, which is originally defined in terms of distributions, by means of continuous functions and brings the theory of de Branges spaces into the analysis of the Weil quadratic form via the Fourier transform. Based on this framework, we formulate a conjecture stating that a self-adjoint operator whose eigenvalues are the imaginary parts of the nontrivial zeros of the Riemann zeta function can be obtained as the limit, as $a \to \infty$, of self-adjoint operators arising from nonlocal realizations of the first-order differential operator on the finite interval $[-a, a]$. All these results are obtained without assuming the Riemann Hypothesis. This conjecture may be compared with the limit formula for the Riemann zeta function expressed in terms of zeta-regularized products proposed by Connes, Consani, and Moscovici, and it sheds new light on the spectral-theoretic interpretation of the nontrivial zeros of the Riemann zeta function.

1. Introduction

1.1. **Historical Background and Motivation.** For test functions f, the Weil functional $f \mapsto W(f)$ is defined by

$$W(f) := \int_{-\infty}^{\infty} f(x)(e^{x/2} + e^{-x/2})dx - \sum_{n=1}^{\infty} \frac{\Lambda(n)}{\sqrt{n}} f(\log n) - \sum_{n=1}^{\infty} \frac{\Lambda(n)}{\sqrt{n}} f(-\log n)$$
$$- (\log 4\pi + C_0)f(0) - \int_0^{\infty} \left\{ f(x) + f(-x) - 2e^{-x/2} f(0) \right\} \frac{e^{x/2}dx}{e^x - e^{-x}},$$

where $\Lambda(n)$ denotes the von Mangoldt function, defined as $\Lambda(n) = \log p$ if $n = p^k$ with $k \in \mathbb{Z}_{>0}$, and $\Lambda(n) = 0$ otherwise, and C_0 is the Euler–Mascheroni constant. Although the precise class of test functions will be specified as needed later, our primary object of interest is the symmetric (or Hermitian) quadratic form

$$Q_W(v_1, v_2) := W(v_1 * \widetilde{v_2}), \quad Q_W(v) := Q_W(v, v),$$

where

$$(v_1 * v_2)(x) := \int_{-\infty}^{\infty} v_1(y) v_2(x-y)\, dy \quad \text{and} \quad \widetilde{v}(x) := \overline{v(-x)}.$$

The Riemann Hypothesis (RH) states that all nontrivial zeros of the Riemann zeta function $\zeta(s)$ lie on the critical line $\Re(s) = 1/2$. A fundamental result due to Weil [15] states that RH is equivalent to the condition that

$$Q_W(v) \geq 0 \quad \text{for all } v \in C_c^{\infty}(\mathbb{R}),$$

a property known as Weil's positivity criterion for RH (although Weil did not formulate the criterion in terms of compactly supported smooth test functions; see [15, 16] and

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[13, Section 3.2]). In the commentary to his Collected Works, Weil emphasized the distributional nature of this criterion and suggested that it deserved further investigation by analysts. Despite this invitation, the functional W and the quadratic form Q_W have not been extensively studied, with notable exceptions such as the works of Yoshida [17], Bombieri [1, 2], Connes and Consani [3], and Connes, Consani, and Moscovici [4].

The study of the localization of the Weil quadratic form Q_W was pioneered by Yoshida [17], who considered the positive definiteness of Q_W on the spaces

$$C_c^\infty(-a,a) := \{v \in C_c^\infty(\mathbb{R}) \mid \mathrm{supp}(v) \subset [-a,a]\}$$

and observed that RH is equivalent to its positive definiteness for every $a > 0$. In [17, Proposition 1], Yoshida further demonstrated that the condition $Q_W(v) \geq 0$ for all non-zero odd functions $v \in C_c^\infty(\mathbb{R})$ implies RH, while a similar condition for all non-zero even functions implies RH except for possible real zeros. Crucially, he initiated the variational study of Q_W by considering the infimum of the Rayleigh quotient $Q_W(v)/\|v\|_{L^2}^2$ on the localized space

$$K(a) := \{v \mid v(x) = f(x)\mathbf{1}_{[-a,a]}(x) \text{ for some } f \in C^\infty(\mathbb{R}) \text{ which has period } 2a\},$$

proving that it is positive for sufficiently small $a > 0$ [17, Lemma 2] or for appropriate subspaces $K_N(a)$ of finite codimension [17, Lemma 3]. The non-degeneracy of Q_W on the "completion" $\widehat{K(a)}$ ($\hookrightarrow L^2(-a,a)$) was shown to be equivalent to RH in [17, Theorem 2]. Note that the hat in $\widehat{K(a)}$ does not denote the Fourier transform.

Building on this foundation, Bombieri [1, 2] addressed the minimization of the Rayleigh quotient on the Sobolev space $H_0^1(-a,a)$ [1, Problem 1] (and [2, Problem B]) and on $L^2(\mathcal{E})$ for a finite union of intervals $\mathcal{E}$ [1, Problem 2] (and [2, Problem A]). While the existence of a minimizer for the lower bound on $L^2(\mathcal{E})$ was established in [1, Theorem 3] (and [2, Theorem 4.3]), the subsequent argument concerning the continuity of this lower bound with respect to a remains an analytically delicate issue. Specifically, although it was claimed in [1, Theorem 5] that the lower bound varies continuously when the minimization is restricted to the subspaces of either even or odd functions, this assertion calls for a more careful analysis. In view of Yoshida's result, such continuity would essentially reproduce the equivalence between the failure of RH and the existence of a degenerating element $v \in \mathfrak{D}(Q_W)$ satisfying $Q_W(v) = 0$, offering a simplification of the original proof that relied on explicit formulas. See the discussion following Theorem 1.3 for further details.

More recently, a rigorous operator-theoretic refinement has been provided by Connes and Consani [3] and Connes, Consani, and Moscovici [4]. They established that the localized closed symmetric form

$$Q_W^a := Q_W|_{L^2(-a,a)}$$

is lower bounded and lower semicontinuous [3, Section 2], leading to the construction of a canonical, densely defined self-adjoint operator A_a on $L^2(-a,a)$ such that

$$Q_W^a(v) = \langle A_a v, v\rangle_{L^2} \tag{1.1}$$

for $v \in \mathfrak{D}(A_a) \subset \mathfrak{D}(Q_W^a) \subset L^2(-a,a)$ [4, Section 3] (here we reformulate their results, originally stated on $[1/\lambda, \lambda]$, in terms of the variable $a = \log\lambda$), where the bracket is the standard L^2 inner product

$$\langle v_1, v_2\rangle_{L^2} = \int_{-\infty}^{\infty} v_1(x)\overline{v_2(x)}dx.$$

This operator A_a, which can be viewed as a refinement of Bombieri's Lagrangian ([1, Lemma 1], [2, Sections 5–6]), possesses a discrete lower bounded spectrum [4, Theorem 3.6] and a corresponding ground state [4, Corollary 3.7]. Furthermore, it was shown in [4, Theorem 5.10] that, when Q_W^a is restricted to a certain finite-dimensional subspace of $L^2(-a,a)$, the Fourier transform of the eigenfunction corresponding to the lowest eigenvalue possesses only real zeros.

First, we aim to unify these various results through the framework of the screw function associated with $\zeta(s)$, which was introduced in [13]. This approach not only provides transparent re-derivations of previous results, but also leads to more refined results presented below. A key advantage of this screw function approach is that it enables us to understand Weil's quadratic form through continuous (and well-behaved) functions, thereby avoiding the delicate distribution-theoretic aspects of Weil's original formulation. The connection between the Weil quadratic form and the screw function, already emphasized in [13, Proposition 3.1], was expected to provide new insight into the study of the Weil quadratic form. Indeed, [13] reinterpreted some of the results by Yoshida [17], Bombieri [1, 2], and Connes–Consani [3] from the viewpoint of screw functions. However, these earlier works had not yet been placed within a fully unified framework. Since then, the theoretical foundations of the screw-function approach to zeta functions have been substantially developed.

In particular, [14] established a completely new picture of the Hilbert space obtained as the completion of $C_c^\infty(\mathbb{R})$ with respect to Q_W under RH. This was made possible by the favorable analytic properties of the screw function as an integral kernel, such as positivity and smoothness. Our main objective is to understand what new insights about Q_W can be obtained by localizing the global framework of [14] to the interval $[-a,a]$, thereby returning to Yoshida's original viewpoint. This line of investigation was further motivated by recent results in [4, Theorem 5.10, Sections 7–8]. Let

$$\xi(s) := s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$$

be the Riemann ξ-function, where $\Gamma(s)$ is the usual gamma function. Let v_a be the eigenfunction corresponding to the lowest eigenvalue of A_a (denoted by ξ_λ with $a = \log\lambda$ in their notation). Denote by

$$\widehat{f}(z) := \int_{-\infty}^{\infty} f(x)\, e^{izx} dx$$

the Fourier transform of f. Motivated by the Hilbert–Pólya operator constructed in [14, Section 6], we next formulate an analogue of the conjectural limiting formula discussed in [4, Section 7],

$$\lim_{a\to\infty} c_a \widehat{v_a}(z) = \xi(1/2+iz). \tag{1.2}$$

This line of investigation led to a unified perspective on the earlier works mentioned above and ultimately to Corollary 1.6, which constitutes the main conjectural statement of the present paper.

1.2. **Main Results.** Before presenting the main results, we emphasize that none of them depends on RH. The first step in the study of the Weil quadratic form via screw functions is to obtain an explicit description of the self-adjoint operator A_a in (1.1). While the existence of A_a was established abstractly in [4, Section 3] via the theory of quadratic forms, the use of screw functions provides a constructive and explicit description of A_a.

Formally, A_a may be viewed as an integral operator with difference kernel $k(x-y)$, where $k(t) = \sum_{\xi(1/2+i\gamma)=0} e^{-i\gamma t}$ is understood as a distribution. Integrating k twice leads to a continuous function. Applying Weil's explicit formula to it yields the following continuous function.

Let $g(t)$ be a continuous real-valued even function on $\mathbb{R}$, defined by

$$\begin{aligned} g(t) = -4(e^{t/2} + e^{-t/2} - 2) + \sum_{n \le \exp(|t|)} \frac{\Lambda(n)}{\sqrt{n}} (|t| - \log n) \\ - \frac{|t|}{2}(\psi(1/4) - \log \pi) - \frac{1}{4}\left(\Phi(1,2,1/4) - e^{-|t|/2}\Phi(e^{-2|t|},2,1/4)\right), \end{aligned} \tag{1.3}$$

where $\psi(s)$ is the digamma function and $\Phi(z,s,a) = \sum_{n=0}^{\infty}(n+a)^{-s}z^n$ is the Hurwitz–Lerch zeta function. A function g on $\mathbb{R}$ satisfying $g(t) = \overline{g(-t)}$ is called a screw function on the real line if the kernel $g(t-u) - g(t) - g(-u) + g(0)$ is nonnegative for all $t, u \in \mathbb{R}$. As shown in [13, Theorem 1.2], the above function g is a screw function in the sense of Krein–Langer [8, Section 5] if and only if RH holds. For this reason, we shall refer to it as the screw function associated with $\zeta(s)$. We define the integral operator G by

$$(Gu)(x) := \int_{-\infty}^{\infty} g(x-y)u(y)\,dy. \tag{1.4}$$

The integral on the right-hand side converges absolutely for $u \in C_c^{\infty}(\mathbb{R})$, since g is continuous. However, as noted immediately after Theorem 1.6 in [13], $g(t)$ does not tend to zero as $|t| \to \infty$, and hence the integral does not necessarily converge for a general $u \in L^2(\mathbb{R})$. Within the theory of screw functions, it would be more natural to consider the integral operator $\widetilde{G}$ with the kernel $g(t-u) - g(t) - g(-u) + g(0)$ (cf. Section 8); nevertheless, for applications to Q_W, the simpler operator G is sufficient.

Throughout this paper, for each $a > 0$, we view $L^2(-a,a)$ as a closed subspace of $L^2(\mathbb{R})$ by extending functions to be zero outside $(-a,a)$. Let $L_0^2(-a,a)$ denote the closed subspace of $L^2(-a,a)$ consisting of functions u satisfying $\widehat{u}(0) = \int_{-a}^{a} u(x)dx = 0$. We denote by P_a the orthogonal projection from $L^2(\mathbb{R})$ onto $L_0^2(-a,a)$ (see (2.1)), and define

$$G_a := P_a G P_a : L_0^2(-a,a) \to L_0^2(-a,a). \tag{1.5}$$

The self-adjointness of the operator G_a follows immediately from the fact that g is real-valued and even. By [13, Proposition 3.1], the integral operator G_a associated with the screw function is directly related to Weil's quadratic form Q_W via differentiation of test functions. Let $D = i\,d/dx$ denote the differential operator on $L^2(-a,a)$ with Dirichlet boundary conditions, and let D^* be its adjoint. We then define

$$B_a := D^* G_a D : L^2(-a,a) \to L^2(-a,a), \quad \mathfrak{D}(B_a) = H_0^1(-a,a). \tag{1.6}$$

This operator is symmetric but fails to be self-adjoint. Then, the operator A_a in (1.1) can be characterized as follows.

Theorem 1.1. *For each $a > 0$, the self-adjoint operator A_a defined by* (1.1) *is the Friedrichs extension of the symmetric operator B_a defined by* (1.6).

This result provides a concrete operator-theoretic realization of the spectral interpretation of Q_W^a established in [4] through the use of screw functions, thereby clarifying the connection between this spectral interpretation and the classical variational approach of Yoshida [17].

Moreover, as shown in [4, Theorem 3.6], for each $a > 0$, the spectrum of the self-adjoint operator A_a is bounded from below and discrete, with $+\infty$ as its only accumulation point. In particular, the largest lower bound of the spectrum of A_a, denoted by λ_a, is an eigenvalue. Therefore, there exists a nonzero function $v \in \mathfrak{D}(A_a)$ attaining the infimum

$$\lambda_a = \inf_{0 \ne v \in \mathfrak{D}(Q_W^a)} \frac{Q_W^a(v)}{\|v\|_{L^2}^2}. \tag{1.7}$$

Furthermore, the domain $\mathfrak{D}(A_a)$ is strictly larger than $\mathfrak{D}(B_a) = H_0^1(-a,a)$ and contains functions such as constants. As will be shown later (Lemma 3.1), we have

$$Q_W^a(v) = \langle B_a v, v\rangle_{L^2} \quad \text{for} \quad v \in H_0^1(-a,a). \tag{1.8}$$

Since the form-norm closure of $C_c^\infty(-a,a)$ ($\subset H_0^1(-a,a)$) with respect to Q_W^a contains the core of the quadratic form Q_W^a as in the proof of Theorem 1.1, we obtain the following result.

Corollary 1.2. *For each $a > 0$, any function attaining the infimum in* (1.7) *belongs to the closure of $C_c^\infty(-a,a)$ with respect to the form norm associated with Q_W^a. In particular, λ_a is the infimum of the Rayleigh quotient*

$$\frac{\langle B_a v, v\rangle_{L^2}}{\|v\|_{L^2}^2}$$

over $C_c^\infty(-a,a)$.

This is analogous to [1, Problem 2] (and [2, Problem A]) and [4, Theorem 3.6], but here the class over which the infimum is taken is made explicit. For the relation between the Rayleigh quotient in [1, Problem 1] (and [2, Problem B]) and the operator G_a, see Section 8.4.

One expects that the eigenvalues of the self-adjoint operator A_a vary continuously in a. However, because A_a is unbounded and its domain is difficult to describe explicitly, it is not straightforward to prove such continuity. Indeed, in [1, Theorem 5] (and [2, Theorem 4.4]), analogues of the infimum in (1.7) are considered on the spaces of even and odd functions, and their continuity with respect to a is asserted, but the details of the proof are not fully provided there.

In contrast, by exploiting the asymptotic expansion (2.2) of the screw function near the origin, which involves a logarithmic singularity, one can prove the continuity of the lowest eigenvalue λ_a without imposing any parity restriction.

Theorem 1.3. *The lowest eigenvalue λ_a is continuous in a.*

Since the continuity of λ_a can be established without assuming RH, Theorem 1.3 immediately yields, as a corollary, another proof of Yoshida's result [17] that RH is equivalent to the nondegeneracy of Q_W^a for every $a > 0$. Indeed, the failure of RH is equivalent to the existence of some $a > 0$ for which $\lambda_a < 0$. Since $\lambda_a > 0$ for sufficiently small $a > 0$ [17, Lemma 2], it follows that if RH is false, then Q_W^a must be degenerate for some value of a by continuity of λ_a.

We now turn to the eigenfunction corresponding to the lowest eigenvalue, following the approach of [4]. Roughly speaking, [4, Theorem 5.10] (see also [4, Section 7]) shows that if the lowest eigenvalue λ_a is simple and its eigenfunction $v_a \in \mathfrak{D}(A_a)$ is even, then all zeros of the Fourier transform $\widehat{v_a}(z)$ (corresponding to $\widehat{\xi_\lambda}(z)$ in [4]) are real. More precisely, they study restrictions of Q_W^a to suitable finite-dimensional subspaces of $L^2(-a,a)$, obtaining the corresponding reality result at each finite-dimensional level. In our framework, however, the simplicity of λ_a and the evenness of the corresponding eigenfunction follow naturally for sufficiently small a by once again exploiting the asymptotic expansion (2.2) of the screw function.

Theorem 1.4. *For sufficiently small $a > 0$, the lowest eigenvalue λ_a is positive, simple, and satisfies*

$$\lambda_a = \log\frac{1}{a} + \mu_1 - \log(2\pi) + \psi(2) - 1 + O(a)$$

as $a \to 0+$, for some constant $\mu_1 > 0$. Furthermore, the corresponding eigenfunction is even.

Let us return to the eigenfunction v_a corresponding to the lowest eigenvalue. The fact that all zeros of its Fourier transform $\widehat{v_a}(z)$ are real follows from the property that it serves as the characteristic function (given by a zeta-regularized product) of a self-adjoint perturbation of the differential operator d/dx with periodic boundary conditions [4, Theorem 5.10]. However, since such a differential operator cannot be defined directly as an operator on $L^2(-a,a)$, the authors of [4] instead consider restrictions of Q_W^a to finite-dimensional subspaces V. Consequently, what is actually established there is not directly the reality of the zeros of $\widehat{v_a}(z)$ itself, but rather the reality of the zeros of the Fourier transform of a function minimizing the Rayleigh quotient of $Q_W^a|_V$.

Instead of considering perturbations of the differential operator d/dx with periodic boundary conditions, we construct an analogue of $\widehat{v_a}(z)$ by considering self-adjoint extensions of the minimal operator $\mathscr{D}_a$ in a Hilbert space different from the usual L^2 space. Choose λ such that $\lambda_a > \lambda$, and define $T_a = T_{a,\lambda} := A_a - \lambda I$. Since T_a is positive,

$$\|v\|_{T_a}^2 := \langle T_a v, v\rangle_{L^2} = Q_W^a(v) - \lambda\|v\|_{L^2}^2 \tag{1.9}$$

defines a norm on $C_c^\infty(-a,a)$. Let $\mathcal{H}(T_a)$ denote the corresponding completion. Then $C_c^\infty(-a,a)$ embeds injectively into $\mathcal{H}(T_a)$. We therefore consider the differential operator $\mathscr{D}_a := i\,d/dx$ on $\mathcal{H}(T_a)$ with domain

$$\mathfrak{D}(\mathscr{D}_a) := C_c^\infty(-a,a). \tag{1.10}$$

It turns out that $\mathscr{D}_a$ is a symmetric operator with deficiency indices $(1,1)$ (Lemmas 6.1 and 6.2). Hence $\mathscr{D}_a$ admits a family of self-adjoint extensions $\overline{\mathscr{D}}_{a,\theta}$ parametrized by $\theta \in [0,2\pi)$. A function arising from the boundary form of the adjoint operator $\mathscr{D}_a^*$ provides a counterpart to $\widehat{v_a}(z)$ in [4], as follows.

Theorem 1.5. *Let $a > 0$ and $\theta \in [0,2\pi)$. Let $v_\pm(a,x)$ be eigenfunctions of the adjoint operator $\mathscr{D}_a^*$ corresponding respectively to the eigenvalues $\pm i$, normalized so that $\|v_+(a,\cdot)\|_{T_a} = \|v_-(a,\cdot)\|_{T_a}$. Then the function*

$$W(a,\theta;z) := (z-i)\int_{-a}^{a} v_+(a,x)e^{izx}dx + e^{i\theta}(z+i)\int_{-a}^{a} v_-(a,x)e^{izx}dx \tag{1.11}$$

is entire in z. The eigenvalues of the self-adjoint operator $\overline{\mathscr{D}}_{a,\theta}$ are precisely the zeros of $W(a,\theta;z)$. Furthermore, all zeros of $W(a,\theta;z)$ are real.

The equation $W(a,\theta;z) = 0$ is equivalent to

$$e^{i\theta} = -\left((z-i)\int_{-a}^{a} v_+(x)e^{izx}\,dx\right)\Big/\left((z+i)\int_{-a}^{a} v_-(x)e^{izx}\,dx\right).$$

This is analogous to the equation $e^{i\theta} = \exp(iza)/\exp(-iza)$, which characterizes the eigenvalues of the self-adjoint extension ∂_θ of the minimal operator $\partial = i\,d/dx$ on $L^2(-a,a)$ through the boundary values of its eigenfunctions.

Let us compare Theorem 1.5 with [4, Theorem 5.10]. First, the two results are parallel in that both establish the reality of the zeros of a certain function through its relation to the self-adjointness of a differential operator. However, whereas the result in [4] requires strong assumptions, such as the simplicity of λ_a and the evenness of the corresponding eigenfunction, our Theorem 1.5 has the advantage of being proved unconditionally. In fact, its proof does not require detailed information on the arithmetic terms appearing in the Weil form Q_W^a. It relies essentially only on the fact that the prime contribution to Q_W^a involves only finitely many primes for fixed $a > 0$. Moreover, they obtain a determinant representation of $\widehat{v_a}(z)$ in terms of a zeta-regularized determinant (by restricting to finite-dimensional subspaces), thereby interpreting $\widehat{v_a}(z)$ as a characteristic function. By contrast, our expression (1.11) arises from a Lagrangian condition.

A further difference appears in the approach to RH, as discussed in [4, Sections 7 and 8]. While their function $\widehat{v_a}(z)$ is expected to approximate $\xi(1/2-iz)$ in the limit $a\to\infty$ through the conjectural formula (1.2), our function $W(a,\theta;z)$ is expected, as explained below, to approximate the reciprocal of one plus its logarithmic derivative.

Corollary 1.6. *If one can choose $\theta = \theta(a)$ and $\phi(a,z)$ ($\neq \infty$ for any $a > 0$ and $z \in \mathbb{C}$) such that*

$$\lim_{a\to\infty} e^{\phi(a,z)}\, W(a,\theta;z) = \frac{\xi(1/2 - iz)}{\xi(1/2 - iz) + \xi'(1/2 - iz)} \tag{1.12}$$

holds uniformly on every compact subset $K \subset \mathbb{C}$, then RH holds.

A direct computation of the eigenfunctions shows that every eigenvalue of $\overline{\mathscr{D}}_{a,\theta}$ has multiplicity one. The right-hand side of (1.12) is consistent with this simplicity of the spectrum. The exponential factor $\exp(\phi(a,z))$ is included to allow for a possible normalization in the limit process. It is plausible that the form of the right-hand side of (1.12) suggests that $\theta = \pi$ may be the most natural choice, but we do not pursue these questions here. The reason why the limit formula (1.12) is expected to hold will be discussed in Section 7. In brief, it is motivated by the determination of the structure of $\mathcal{H}(A_\infty) := \mathcal{H}(T_{\infty,0})$ under RH in [14, Theorem 1.1], together with the construction of the Hilbert–Pólya operator described in [14, Section 6].

Under RH, one has $A_a > 0$, so that $\lambda = 0$ may be chosen in T_a for every $a > 0$. The limit formula (1.12) is formulated with the situation $T_a = A_a$ (i.e. $\lambda = 0$) in mind, and indeed Section 7 proceeds under the assumption $A_a > 0$. Nevertheless, when $A_a > 0$, both $\mathcal{H}(T_a)$ and $\mathcal{H}(A_a)$ are defined and are isomorphic as Hilbert spaces. Hence the zeros of $W(a,\theta;z)$ are expected to be independent of the choice of λ used in the definition of T_a. However, in any attempt to prove the limit formula (1.12), control of λ ($< \lambda_a$) will likely play an important role. Such control is expected to require a detailed analysis of the arithmetic contribution coming from the prime terms in Q_W^a, which lies beyond the arguments used in the proof of Theorem 1.5, where only the finiteness of the prime contribution for fixed a plays a role.

Since the reality of the zeros of $W(a,\theta;z)$ for each $a > 0$ has already been established in Theorem 1.5, it would suffice to prove (1.12) as an identity of complex functions in order to deduce RH. As explained in Section 8, the eigenfunctions $v_\pm(a,x)$ are solutions of Fredholm integral equations of the first kind associated with the integral operator G_a whose kernel is the continuous function $g(x-y)$. Consequently, the conjectural limit formula (1.12) in Corollary 1.6 can be formulated as a purely analytic statement, independently of $\mathcal{H}(T_a)$.

The organization of this paper is as follows. In Section 2, we study the asymptotic expansion of the screw function g near the origin and derive from it several formulas for the quadratic form $\langle B_a v, v\rangle$. The latter formulas will be used repeatedly in the proofs of the main results. Using the results of Section 2, we prove Theorem 1.1 in Section 3 and Theorem 1.3 in Section 4. In both proofs, the compactness of the embedding $\mathfrak{D}(Q_W^a) \to L^2(-a,a)$ plays an essential role. Theorem 1.4 is proved in Section 5. In addition to the results of Section 2, the proof relies on the theory of Dirichlet forms. In Section 6, we prove Theorem 1.5 using von Neumann's theory of self-adjoint extensions. Section 7 explains the motivation for the limit formula in Corollary 1.6 by relating it to the results of [14]. The main tool there is the theory of de Branges spaces. Finally, in Section 8, we relate the theory of the quadratic forms $Q_W^a(v) = \langle A_a v, v\rangle_{L^2}$ developed in this paper to the theory of the quadratic forms $\langle G_a u, u\rangle_{L^2}$. While the former belongs to the theory of distributions and is analytically more delicate, the latter has the advantage of being treatable within the theory of integral operators with continuous kernels.

2. Preliminaries

2.1. **Precise definition of operators.** Let

$$L_0^2(-a,a) = \{u \in L^2(-a,a) \mid \widehat{u}(0) = 0\},$$

as in Section 1. Then $D = i\,d/dx$ denotes the differentiation operator on $L^2(-a,a)$ with Dirichlet boundary conditions $v(\pm a) = 0$, whose domain and range are $\mathfrak{D}(D) =$

$H_0^1(-a,a)$ and $\mathfrak{R}(D) = L_0^2(-a,a)$, respectively. Let D^* denote the adjoint of D with respect to the standard L^2 inner product. While D^* acts as the same differential operator $i\,d/dx$, its domain is $\mathfrak{D}(D^*) = H^1(-a,a)$ and its range is $\mathfrak{R}(D^*) = L^2(-a,a)$. Consequently, D is not self-adjoint. Define the projection $P_a : L^2(\mathbb{R}) \to L_0^2(-a,a)$ by

$$(P_a u)(x) := u(x) - \frac{1}{2a}\widehat{u}(0) = u(x) - \frac{1}{2a}\int_{-a}^{a} u(y)\,dy \tag{2.1}$$

for $x \in (-a,a)$, and $(P_a u)(x) = 0$ otherwise. Then,

$$(P_a K u)(x) = (Ku)(x) - \frac{1}{2a}\int_{-a}^{a} (Ku)(t)\,dt, \quad x \in (-a,a).$$

This defines the operator G_a in (1.5). The kernel function g is continuous, and its first derivative is piecewise continuous with only a discrete set of discontinuities at which finite one-sided limits exist. It follows that for any $u \in L_0^2(-a,a)$, we have $G_a u \in \mathfrak{D}(D^*) = H^1(-a,a)$. Consequently, the operator B_a on $L^2(-a,a)$ with domain $\mathfrak{D}(B_a) := H_0^1(-a,a)$ is well-defined by (1.6). Since G_a is self-adjoint, the relation $\langle B_a v, w\rangle_{L^2} = \langle v, B_a w\rangle_{L^2}$ holds for all $v, w \in \mathfrak{D}(B_a)$. Thus, B_a is a symmetric operator, but it is not self-adjoint. Indeed, the domain of the adjoint operator, $\mathfrak{D}(B_a^*)$, consists of all w such that the functional $v \mapsto \langle B_a v, w\rangle$ is L^2-continuous, which also includes functions $w \in H^1(-a,a)$ that do not vanish on the boundary. That is, $\mathfrak{D}(B_a) \subsetneq \mathfrak{D}(B_a^*)$.

2.2. **Expansion near the origin.** We derive an asymptotic formula for the screw function g in a neighborhood of the origin. The resulting asymptotic formula will be used repeatedly throughout the paper. [6, 9.554] with $z = e^{-2t}$, $m = 2$, and $v = 1/4$ yields

$$\begin{aligned} e^{-t/2}\Phi(e^{-2t}, 2, 1/4) &= 2t(\log(2t) + \psi(1/4) - \psi(2)) \\ &\quad + \zeta(2, 1/4) + \sum_{n=2}^{\infty} \zeta(2-n, 1/4)\frac{(-2t)^n}{n!} \end{aligned}$$

for $t \to 0+$, where $\zeta(s,a)$ denotes the Hurwitz zeta function. By definition (1.3),

$$\begin{aligned} g(t) = {}& -4(e^{t/2} + e^{-t/2} - 2) - (|t|/2)(\psi(1/4) - \log\pi) - (1/4)(F(0) - F(t)) \\ & + \sum_{n \le \exp(|t|)} \frac{\Lambda(n)}{\sqrt{n}}(|t| - \log n), \end{aligned}$$

where

$$F(t) := 2|t|(\log(2|t|) + \psi(1/4) - \psi(2)) + \zeta(2, 1/4) + \sum_{n=2}^{\infty}\zeta(2-n, 1/4)\frac{(-2|t|)^n}{n!}.$$

We have $\psi(2) = 1 - C_0$ [6, (8.365), (8.366)] and $\psi(1/4) = -(\pi/2) - 3\log 2 - C_0$ [6, (8.366)], where C_0 denotes the Euler–Mascheroni constant. It follows from the defining series that $\Phi(1, 2, 1/4) = \zeta(2, 1/4)$. Using also $\zeta(0, 1/4) = 1/4$ [6, (9.523)], we obtain

$$g(t) = \frac{1}{2}|t|\log|t| + A|t| + \sum_{n \le \exp(|t|)} \frac{\Lambda(n)}{\sqrt{n}}(|t| - \log n) + r(t) \tag{2.2}$$

in a neighbourhood of $t = 0$, where r is an even C^2-function satisfying $r(t) = O(t^2)$, and

$$A = \frac{1}{2}(\log(2\pi) - \psi(2)) = \frac{1}{2}(\log(2\pi) + C_0 - 1) = 0.707546\ldots.$$

The sum involving the von Mangoldt function in (2.2) is understood to be zero if $|t| < (1/2)\log 2$.

2.3. **Quadratic form formulas.** For the quadratic form

$$Q_{B_a}(v) := \langle B_a v, v\rangle_{L^2}, \quad v \in \mathfrak{D}(B_a) = H_0^1(-a,a)$$

we compute the quadratic forms corresponding to each term on the right-hand side of (2.2). First, for the first term on the right-hand side, we have

$$\int_{-a}^{a}\int_{-a}^{a} |x-y| v'(y)\overline{v'(x)}dxdy = -2\|v\|_{L^2}^2, \quad v \in \mathfrak{D}(B_a).$$

Next, for the second term on the right-hand side, if we define

$$\mathcal{L}_a(v) := \frac{1}{4}\int_{-a}^{a}\int_{-a}^{a} \frac{|v(x)-v(y)|^2}{|x-y|}\,dx\,dy - \frac{1}{2}\int_{-a}^{a} \log(a^2-x^2)|v(x)|^2\,dx \tag{2.3}$$

for $v \in \mathfrak{D}(B_a)$ (to ensure the convergence of the second term), then we obtain

$$\int_{-a}^{a}\int_{-a}^{a} |x-y|\log|x-y| v'(y)\overline{v'(x)}dxdy = 2\mathcal{L}_a(v) - 2\|v\|_{L^2}^2, \quad v \in \mathfrak{D}(B_a). \tag{2.4}$$

We verify this relation. Let $I(v)$ denote the left-hand side. Applying integration by parts with respect to y, we have

$$\int_{-a}^{a} |x-y|\log|x-y| v'(y)dy = -\int_{-a}^{a} \operatorname{sgn}(y-x)(\log|x-y|+1)v(y)dy.$$

Substituting this into $I(v)$, we introduce an ϵ-cutoff to handle the singularity of the kernel rigorously:

$$I(v) = -\lim_{\epsilon\to 0}\int_{-a}^{a} \overline{v'(x)}\left(\int_{|y-x|>\epsilon} \operatorname{sgn}(y-x)(\log|x-y|+1)v(y)dy\right)dx.$$

Let $F_\epsilon(x)$ denote the inner integral inside the brackets. Differentiating it by Leibniz's rule, we obtain

$$F_\epsilon'(x) = -(\log\epsilon+1)v(x-\epsilon) + v(x+\epsilon) - \int_{|y-x|>\epsilon} \frac{v(y)}{|x-y|}dy.$$

Using this, we integrate by parts with respect to x. Taking into account the boundary conditions $v(\pm a) = 0$ and the fact that $v(x\pm\epsilon) \to v(x)$ as $\epsilon \to 0$, we find

$$I(v) = -\lim_{\epsilon\to 0}\int_{-a}^{a}\left[2(\log\epsilon+1)|v(x)|^2 + \int_{|y-x|>\epsilon} \frac{\overline{v(x)}v(y)}{|x-y|}dy\right]dx.$$

Here, we use the identity $2\mathrm{Re}(\overline{v(x)}v(y)) = |v(x)|^2 + |v(y)|^2 - |v(x)-v(y)|^2$ to convert the product into a difference of squares (the imaginary part vanishes due to the symmetry of the double integral). Evaluating the integral concerning $|v(x)|^2$ yields

$$\int_{y\in(-a,a),|y-x|>\epsilon} \frac{1}{|x-y|}dy = -2\log\epsilon + \log(a^2-x^2).$$

Substituting this and simplifying the coefficient of $|v(x)|^2$, the contribution from the first term is $2\log\epsilon + 2$, while the contribution from the second term is $-2\log\epsilon + \log(a^2-x^2)$. Consequently, $\log\epsilon$ is cancelled out, and we obtain

$$I(v) = -\int_{-a}^{a} |v(x)|^2\left(2+\log(a^2-x^2)\right)dx + \frac{1}{2}\int_{-a}^{a}\int_{-a}^{a} \frac{|v(x)-v(y)|^2}{|x-y|}dxdy.$$

Rearranging the terms yields (2.4).

Furthermore, setting $g_0(t) = \sum_{n\le\exp(|t|)} n^{-1/2}\Lambda(n)\big(|t| - \log n\big)$ and calculating the corresponding integral,

$$\begin{aligned}
&\int_{-a}^{a}\int_{-a}^{a} g_0(x-y)\, v'(y)\overline{v'(x)}\, dx\, dy \\
&= -\sum_{n\le\exp(2a)} \frac{\Lambda(n)}{\sqrt{n}}\left(\int_{-a}^{a-\log n} v(x+\log n)\overline{v(x)}\, dx + \int_{-a+\log n}^{a} v(x-\log n)\overline{v(x)}\, dx\right).
\end{aligned}$$

Computing $Q_{B_a}(v)$ by using the asymptotic expansion (2.2), we get

$$\begin{aligned}
Q_{B_a}(v) = \mathcal{L}_a(v) &- (2A+1)\|v\|_{L^2}^2 \\
&- \sum_{n\le\exp(2a)} \frac{\Lambda(n)}{\sqrt{n}}\left(\int_{-a}^{a-\log n} v(x+\log n)\overline{v(x)}\, dx\right. \\
&\qquad\qquad \left. + \int_{-a+\log n}^{a} v(x-\log n)\overline{v(x)}\, dx\right) \\
&- \int_{-a}^{a}\int_{-a}^{a} r''(x-y)\, v(y)\overline{v(x)} dx dy
\end{aligned} \tag{2.5}$$

for $v \in H_0^1(-a,a)$.

2.4. **Formulas in Fourier transforms.** Expanding the numerator of

$$\frac{1}{4}\iint_{(-a,a)^2, |x-y|>\epsilon} \frac{|v(x)-v(y)|^2}{|x-y|}\, dx\, dy.$$

as $|v(x)|^2 + |v(y)|^2 - 2\operatorname{Re}(v(x)\overline{v(y)})$ and using symmetry, it equals

$$\frac{1}{2}\int_{-a}^{a} |v(x)|^2 \left(\int_{(-a,a)\setminus(x-\epsilon,x+\epsilon)} \frac{1}{|x-y|}\, dy\right) dx - \frac{1}{2}\iint_{(-a,a)^2,\ |x-y|>\epsilon} \frac{v(x)\overline{v(y)}}{|x-y|}\, dx\, dy.$$

Since

$$\int_{-a}^{x-\epsilon} \frac{1}{x-y}\, dy + \int_{x+\epsilon}^{a} \frac{1}{y-x}\, dy = \log(a^2 - x^2) - 2\log\epsilon,$$

we obtain

$$\mathcal{L}_a(v) = \lim_{\epsilon\to 0}\left[-\log\epsilon \int_{-a}^{a} |v(x)|^2\, dx - \frac{1}{2}\iint_{(-a,a)^2,\ |x-y|>\epsilon} \frac{v(x)\overline{v(y)}}{|x-y|}\, dx\, dy\right].$$

The second term may be rewritten, by Plancherel's theorem, as the quadratic form associated with the convolution kernel $K_\epsilon(x) = |x|^{-1}\mathbf{1}_{\{|x|>\epsilon\}}(x)$. Its Fourier transform is

$$\widehat{K_\epsilon}(z) = \int_{|x|>\epsilon} \frac{e^{-izx}}{|x|}\, dx = 2\int_{\epsilon|z|}^{\infty} \frac{\cos u}{u}\, du.$$

Using the asymptotic expansion of the cosine integral [6, (8.230)], we obtain

$$\int_{\epsilon|z|}^{\infty} \frac{\cos u}{u}\, du = -\log\epsilon - \log|z| - C_0 + o(1) \qquad (\epsilon \to 0).$$

Substituting this into the Fourier representation yields

$$\begin{aligned}
\mathcal{L}_a(v) &= \lim_{\epsilon\to 0} \frac{1}{4\pi}\int_{-\infty}^{\infty} \big(-2\log\epsilon - \widehat{K_\epsilon}(z)\big)|\widehat{v}(z)|^2\, dz \\
&= \frac{1}{2\pi}\int_{-\infty}^{\infty} \big(\log|z| + C_0\big)|\widehat{v}(z)|^2\, dz.
\end{aligned} \tag{2.6}$$

Using the Fourier transform $\widehat{v}$ of v (defined as the zero-extension of v outside $(-a,a)$)

$$\int_{-\infty}^{\infty} v(x \mp \tau)\overline{v(x)}dx = \frac{1}{2\pi}\int_{-\infty}^{\infty} |\widehat{v}(z)|^2 e^{\pm iz\tau}dz,$$

which yields

$$\begin{aligned}\int_{-a}^{a-\log n} v(x+\log n)\overline{v(x)}\,dx + \int_{-a+\log n}^{a} v(x-\log n)\overline{v(x)}\,dx \\ = \frac{1}{\pi}\int_{-\infty}^{\infty} |\widehat{v}(z)|^2 \cos(z\log n)dz.\end{aligned}$$

Furthermore, noting that $\|v\|_{L^2}^2 = (2\pi)^{-1}\|\widehat{v}\|_{L^2}^2$, we obtain from (2.5) that

$$\begin{aligned}Q_{B_a}(v) = {} & \frac{1}{2\pi}\int_{-\infty}^{\infty} (\log|z| - \log(2\pi))\,|\widehat{v}(z)|^2 dz \\ & - \frac{1}{2\pi}\int_{-\infty}^{\infty} |\hat{v}(z)|^2 \left[\sum_{n\le \exp(2a)} \frac{\Lambda(n)}{\sqrt{n}} 2\cos(z\log n)\right] dz \\ & - \frac{1}{2\pi}\int_{-\infty}^{\infty} \widehat{r_a''}(z)|\widehat{v}(z)|^2 dz\end{aligned} \tag{2.7}$$

for $v \in H_0^1(-a,a)$. To interpret the term $\widehat{r_a''}(z)$, we decompose $r(t)$ as $r(t) = r_0(t) + r_1(t)$, where

$$r_0(t) = -4(e^{t/2} + e^{-t/2} - 2) \quad \text{and} \quad r_1(t) = \frac{1}{4}\sum_{n=2}^{\infty} \zeta(2-n, 1/4)\frac{(-2|t|)^n}{n!}.$$

Since $\widehat{r''}(z)$ cannot be defined directly, we define $\widehat{r_a''}(z)$ by treating these two parts separately. For r_0, we take the Fourier transform of $r_{0,a}''(x) := r_0''(x)\mathbf{1}_{(-2a,2a)}(x)$, whereas for r_1 we compute $\widehat{r_1''}(z)$ explicitly. We then set $\widehat{r_a''}(z) := \widehat{r_{0,a}''}(z) + \widehat{r_1''}(z)$. As shown below, all resulting terms except those arising from the polar factor $s(s-1)$ coincide with the explicit formula.

Using $\zeta(2-n,a) = -B_{n-1}(a)/(n-1)$ [10, (25.11.14)] and the generating series $ye^{ay}/(e^y-1) = \sum_{n=0}^{\infty} B_n(a)y^n/n!$ [6, (9.621)], we have

$$r_1''(t) = \frac{e^{-|t|/2}}{1-e^{-2|t|}} - \frac{1}{2|t|}.$$

Hence, the Fourier transform is given by

$$\begin{aligned}\widehat{r_1''}(z) &= 2\int_0^{\infty}\left(\frac{e^{-t/2}}{1-e^{-2t}} - \frac{1}{2t}\right)\cos(zt)\,dt = \int_0^{\infty}\left(\frac{e^{-u/4}}{1-e^{-u}} - \frac{1}{u}\right)\cos(zu/2)\,du \\ &= \lim_{\substack{\epsilon\to+0 \\ R\to\infty}} \left[\int_{\epsilon}^{R}\left(\frac{e^{-u/4}\cos(zu/2)}{1-e^{-u}} - \frac{e^{-u}}{u}\right)du + \int_{\epsilon}^{R} \frac{e^{-u} - \cos(zu/2)}{u}\,du\right].\end{aligned}$$

Differentiating [6, (8.341.3)] with respect to z, we obtain

$$\psi(s) = \int_0^{\infty}\left(\frac{e^{-u}}{u} - \frac{e^{-su}}{1-e^{-u}}\right)du \quad (\mathrm{Re}(s) > 0).$$

By taking the real part for $s = 1/4 + iz/2$, we find

$$\int_0^{\infty}\left(\frac{e^{-u/4}\cos(zu/2)}{1-e^{-u}} - \frac{e^{-u}}{u}\right)du = -\,\mathrm{Re}\left[\psi\left(\frac{1}{4} + \frac{iz}{2}\right)\right]$$

for real z. On the other hand, we have

$$\int_\epsilon^\infty \frac{e^{-u}}{u}\,du - \int_\epsilon^\infty \frac{\cos(zu/2)}{u}\,du = \mathrm{E}_1(\epsilon) + \mathrm{Ci}(|z/2|\epsilon) = \log|z| - \log 2 + O(\epsilon)$$

for real z by $\mathrm{E}_1(\epsilon) = -C_0 - \log\epsilon + O(\epsilon)$ [10, (6.6.2)] and $\mathrm{Ci}(z) = C_0 + \log z + O(|z|)$ [10, (6.6.6)]. Therefore,

$$\widehat{r_1''}(z) = -\operatorname{Re}\left[\psi\left(\frac{1}{4} + \frac{iz}{2}\right)\right] + \log|z| - \log 2 = O(|z|^{-2}) \quad |z| \to \infty.$$

It follows that

$$\begin{aligned} Q_{B_a}(v) &= \frac{1}{2\pi}\int_{-\infty}^{\infty}\left(\operatorname{Re}\left[\psi\left(\frac{1}{4}+\frac{iz}{2}\right)\right] - \log\pi\right)|\widehat{v}(z)|^2 dz \\ &\quad - \frac{1}{2\pi}\int_{-\infty}^{\infty}|\hat{v}(z)|^2\left[\sum_{n\le\exp(2a)}\frac{\Lambda(n)}{\sqrt{n}}2\cos(z\log n)\right]dz \\ &\quad - \frac{1}{2\pi}\int_{-\infty}^{\infty}\widehat{r_{0,a}''}(z)|\widehat{v}(z)|^2 dz. \end{aligned}$$

This formula precisely reflects the structure of the explicit formula. From the explicit formula [1, Lemma 3 and its proof], we have

$$\begin{aligned} Q_W^a(v) = \frac{1}{2\pi}\int_{-\infty}^{\infty}\Re\left[\psi\left(\frac{1}{4}+\frac{iz}{2}\right)\right]|\widehat{v}(z)|^2 dz - \log\pi\int_{-a}^{a}|v(x)|^2 dx \\ + O(a)\cdot\|v\|_{L^2}^2, \end{aligned}$$

where $O(a)$ denotes a bounded quantity depending on a. Since

$$\operatorname{Re}[\psi(s/2)] = \log|s| - \log 2 + O(|s|^{-2}) \quad (\Re(s) > 0) \quad |s| \to \infty,$$

we obtain

$$Q_W^a(v) = \frac{1}{2\pi}\int_{-\infty}^{\infty}(\log|z| - \log(2\pi))\,|\widehat{v}(z)|^2 dz + O(a)\cdot\|v\|_{L^2}^2. \tag{2.8}$$

This perfectly matches the result obtained from (2.7).

2.5. **Distributional formulas.** Since g is continuous, the distributional second derivative $-g''$ is a tempered distribution. By (1.8) and integration by parts,

$$\begin{aligned} Q_W^a(v) &= \int_{-a}^{a}\int_{-a}^{a} g(x-y)v'(y)\overline{v'(x)}\,dx\,dy \\ &= \int_{-a}^{a}\int_{-a}^{a}(-g''(x-y))v(y)\overline{v(x)}\,dx\,dy \end{aligned} \tag{2.9}$$

for $v \in C_c^\infty(-a,a)$. This was already observed in [13, Section 3.5]. Combining this with (1.1), we obtain

$$(A_a v)(x) = \int_{-a}^{a}(-g''(x-y))v(y)\,dy \tag{2.10}$$

for $v \in C_c^\infty(-a,a)$. Since g'' is a tempered distribution, the right-hand side is well-defined. Formula (2.10) provides an alternative representation of Bombieri's Lagrangian ([1, Lemma 1], [2, Sections 5–6]). Let us compute the right-hand side of (2.10). Since

$$\frac{d^2}{dt^2}(|t|\log|t|) = \operatorname{Pf}\left(\frac{1}{|t|}\right) + 2\delta(t)$$

in the sense of distributions, it follows from (2.2) that

$$-g''(t) = -\frac{1}{2}\operatorname{Pf}\left(\frac{1}{|t|}\right) - (2A+1)\delta(t) - \sum_{n\geq 2}\frac{\Lambda(n)}{\sqrt{n}}\big(\delta(t-\log n)+\delta(t+\log n)\big) - r''(t).$$

Hence

$$\begin{aligned}\int_{-a}^{a}(-g''(x-y))v(y)\,dy = -\frac{1}{2}\operatorname{Pf}\int_{-a}^{a}\frac{v(y)}{|x-y|}\,dy - (2A+1)v(x)\\ - \sum_{n\leq e^{2a}}\frac{\Lambda(n)}{\sqrt{n}}\big(v(x-\log n)+v(x+\log n)\big)\\ - \int_{-a}^{a} r''(x-y)v(y)\,dy.\end{aligned} \tag{2.11}$$

Here we note that

$$\begin{aligned}&\int_{-a}^{a}\left(\operatorname{Pf}\int_{-a}^{a}\frac{v(y)}{|x-y|}\,dy\right)\overline{v(x)}\,dx\\ &= -\frac{1}{2}\int_{-a}^{a}\int_{-a}^{a}\frac{|v(x)-v(y)|^2}{|x-y|}\,dx\,dy + \int_{-a}^{a}\log(a^2-x^2)|v(x)|^2\,dx\end{aligned}$$

which follows by a direct symmetrization argument. Using this identity, one verifies that (2.9) follows from (2.5).

For a general element $v \in \mathfrak{D}(A_a) \subset \mathfrak{D}(Q_W^a)$, pointwise values need not be well defined, so an expression such as (2.11) is not meaningful. However, by (2.7), identity (2.9) remains valid for $v \in \mathfrak{D}(A_a)$ in the sense of limits of values on elements of $C_c^\infty(-a,a)$ with respect to the form norm of Q_W^a.

3. Proof of Theorem 1.1

3.1. **The Relation between A_a and B_a.** To compare the operators A_a and B_a, we begin by recalling some basic facts concerning the quadratic form Q_W^a. For general background on quadratic forms, we refer to Schmüdgen [12, Chapter 10].

Let Γ denote the set of all zeros of the function $z \mapsto \xi(1/2 - iz)$. For each $\gamma \in \Gamma$, let m_γ be the multiplicity of γ as a zero of $\xi(1/2-iz)$. By the functional equations $\xi(1-s)=\xi(s)$ and $\xi(s)=\overline{\xi(\bar{s})}$, the set Γ is invariant under $\gamma \mapsto -\gamma$ and $\gamma \mapsto \bar{\gamma}$. It is customary to write the zeros of $\xi(s)$ as $\rho = \beta + i\gamma$ $(\beta,\gamma \in \mathbb{R})$, but our notation differs from this convention. In particular, an element $\gamma \in \Gamma$ may be non-real. RH is equivalent to the inclusion $\Gamma \subset \mathbb{R}$. In this notation, the Weil explicit formula yields

$$Q_W(v_1,v_2) = \sum_{\gamma\in\Gamma} m_\gamma \widehat{v_1}(\gamma)\overline{\widehat{v_2}(\bar{\gamma})} \tag{3.1}$$

(cf. [14, (1.2), (3.3)]). The form domain of the closed symmetric quadratic form Q_W^a is defined by

$$\mathfrak{D}(Q_W^a) := \{\, v \in L^2(-a,a) \mid |Q_W^a(v)| < \infty \,\},$$

and becomes a Hilbert space with respect to the form norm

$$\|v\|_{Q_W^a}^2 := Q_W^a(v) + (1-\lambda_a)\|v\|_{L^2}^2,$$

where λ_a is the constant defined in (1.7). By (2.6) and (2.8), if we define

$$\begin{aligned}H^{\log}(-a,a) :&= \left\{ v \in L^2(-a,a) \;\middle|\; \int_{-\infty}^{\infty}(1+\log^+|z|)|\widehat{v}(z)|^2\,dz < \infty \right\}\\ &\supset \{\, v \in L^2(-a,a) \mid \mathcal{L}_a(v) < \infty \,\},\end{aligned}$$

then

$$\mathfrak{D}(Q_W^a) \subset H^{\log}(-a,a).$$

In [2, Theorem 5.1], Bombieri appears to state implicitly that $\mathfrak{D}(Q_W^a)$ coincides with the set of all $v \in L^2(-a,a)$ such that

$$\int_{-a}^{a}\int_{-a}^{a} \frac{|v(x)-v(y)|^2}{|x-y|}\,dx\,dy + \int_{-a}^{a}|v(x)|^2\,dx$$

is finite. However, taking into account the equivalence between (2.3) and (2.6), one finds that this space is in fact slightly larger than $\mathfrak{D}(Q_W^a)$.

The fact that $Q_W^a(v)$ is lower bounded and lower semicontinuous on $L^2(-a,a)$ was proved in [3, Proposition 2.1]. It is also mentioned in the proof of [1, Lemma 3] and is implicitly contained in [17, Section 2]. All of these arguments are based on the formula (2.8). The lower semicontinuity follows immediately from an application of Fatou's lemma.

We now recall the self-adjoint operator A_a associated with Q_W^a through (1.1). Such a self-adjoint operator is uniquely determined. Moreover, $v \in \mathfrak{D}(A_a) \subset \mathfrak{D}(Q_W^a)$ if and only if the map $w \longmapsto Q_W^a(v,w)$ is continuous on $\mathfrak{D}(Q_W^a)$ with respect to the L^2-norm. By the Riesz representation theorem, this is equivalent to [12, Definition 10.4]. In this case, $A_a v$ is uniquely characterized by $Q_W^a(v,w) = \langle A_a v, w\rangle_{L^2}$ $(w \in \mathfrak{D}(Q_W^a))$.

Lemma 3.1. *$B_a \subset A_a$. In particular, A_a is a self-adjoint extension of B_a.*

Proof. Let $E \subset L^2(-a,a)$ be the linear subspace generated by

$$e_n(x) = \exp(i\pi n x/a), \qquad n \in \mathbb{Z}.$$

The space E is a core for the quadratic form Q_W^a [3, Proposition 2.3]. On the other hand, if $v \in \mathfrak{D}(B_a) = H_0^1(-a,a)$, then $\widehat{v}(z) = O(|z|^{-1})$ as $|z| \to \infty$. Hence, by (3.1), $\mathfrak{D}(B_a) \subset \mathfrak{D}(Q_W^a)$. Moreover,

$$\widehat{e_n}(z) = \frac{2\sin(az+n\pi)}{z+n\pi/a},$$

so that $\widehat{w}(z) = O(|z|^{-1})$ for every $w \in E$. If $u = Dv$, then $\widehat{v}(z) = z^{-1}\widehat{u}(z)$. Therefore, by (3.1) and [13, (3.1)], $Q_W^a(v,w) = \langle B_a v, w\rangle_{L^2}$ for all $v \in H_0^1(-a,a) = \mathfrak{D}(B_a)$ and $w \in E$. It follows from [12, Proposition 10.5 (v)] that $B_a \subset A_a$. $\square$

3.2. **Completion of the proof.** For simplicity, we write $A = A_a$, $B = B_a$, $Q_B(v) = \langle Bv, v\rangle_{L^2}$, $Q = Q_W^a$, and $\|v\|_Q^2 = Q(v) + (1-\lambda_a)\|v\|_{L^2}^2$. Note that $Q = Q_B$ on $\mathfrak{D}(B_a)$. The Friedrichs extension of the operator B is the uniquely determined self-adjoint extension associated with the closure $\overline{Q}_B$ of the quadratic form Q_B. The domain $\mathfrak{D}(\overline{Q}_B)$ of $\overline{Q}_B$ is defined as the completion of the form domain $\mathfrak{D}(Q_B) := \mathfrak{D}(B) = H_0^1(-a,a)$ with respect to the form norm $\|\cdot\|_Q$. Since $B \subset A$ (Lemma 3.1), we have $\mathfrak{D}(\overline{Q}_B) \subset \mathfrak{D}(Q_A) = \mathfrak{D}(Q)$. Hence, $\mathfrak{D}(\overline{Q}_B)$ is a closed subspace of $\mathfrak{D}(Q)$ with respect to the form norm. On the other hand, the space E (in the proof of Lemma 3.1) is a core for the quadratic form Q (i.e. $\overline{E}^{\|\cdot\|_Q} = \mathfrak{D}(Q)$). Since $C_c^\infty(-a,a) \subset H_0^1(-a,a)$, if

$$E \subset \overline{C_c^\infty(-a,a)}^{\|\cdot\|_Q}, \tag{3.2}$$

then $\mathfrak{D}(\overline{Q}_B) = \mathfrak{D}(Q)$. It follows that the Friedrichs extension of B coincides with A. Indeed, by [12, Corollary 10.8], on a given Hilbert space, lower bounded self-adjoint operators are in one-to-one correspondence with lower bounded closed symmetric quadratic forms having dense domains.

To prove the inclusion (3.2), it suffices to show that each function $e_n \in E$ belongs to the closure of $C_c^\infty(-a,a)$ with respect to the form norm $\|\cdot\|_Q$. In other words, we must construct a sequence $v_\epsilon \in C_c^\infty(-a,a)$ such that $\|e_n - v_\epsilon\|_Q \to 0$ as $\epsilon \to 0$. For a sufficiently small parameter $\epsilon > 0$, we define a cutoff function $\eta_\epsilon(x)$ which decays to 0 near the boundary of the interval $[-a, a]$. More precisely, $\eta_\epsilon(x) = 1$ on $[-a+\epsilon, a-\epsilon]$, and $\eta_\epsilon(\pm a) = 0$. On the boundary intervals $[-a, -a+\epsilon]$ and $[a-\epsilon, a]$, η_ϵ is chosen as a smooth monotone cutoff interpolating between these values, with $|\eta'_\epsilon(x)| \ll \epsilon^{-1}$. Define $v_\epsilon(x) := e_n(x)\eta_\epsilon(x)$. Then v_ϵ belongs to $C_c^\infty(-a, a)$. It therefore remains to show that the form norm of the error function $w_\epsilon(x) = e_n(x)(1 - \eta_\epsilon(x))$ tends to zero as $\epsilon \to 0$.

The function w_ϵ is supported only on the two intervals $[-a, -a+\epsilon]$ and $[a-\epsilon, a]$, each of length ϵ. Moreover, $|w_\epsilon(x)| \le 1$. Therefore,

$$\|w_\epsilon\|_{L^2}^2 \le \int_{-a}^{-a+\epsilon} 1\, dx + \int_{a-\epsilon}^{a} 1\, dx = 2\epsilon.$$

Hence $\|w_\epsilon\|_{L^2} \to 0$ as $\epsilon \to 0$.

By (2.7) and Plancherel's theorem, we have

$$\|w_\epsilon\|_Q^2 = \int_{-\infty}^{\infty} (1 + \log^+ |z|)\, |\widehat{w_\epsilon}(z)|^2\, dz + O(a) \cdot \|w_\epsilon\|_{L^2}^2.$$

Therefore, the principal term in $\|w_\epsilon\|_Q^2$ is

$$I_\epsilon = \int_{-\infty}^{\infty} \log(3 + |z|) |\widehat{w_\epsilon}(z)|^2\, dz.$$

To estimate it, we examine the behavior of $\widehat{w_\epsilon}$. First, it follows immediately from the definition of the Fourier transform that

$$|\widehat{w_\epsilon}(z)| \le \int_{\mathbb{R}} |w_\epsilon(x)|\, dx \le 2\epsilon. \tag{3.3}$$

On the other hand, $|\widehat{w_\epsilon}(z)| = O(|z|^{-1})$ as $|z| \to \infty$, uniformly in ϵ. Indeed, integrating by parts in $\widehat{w_\epsilon}(z) = \int_{-a}^{a} w_\epsilon(x) e^{izx}\, dx$, we obtain

$$\widehat{w_\epsilon}(z) = \left[w_\epsilon(x) \frac{e^{izx}}{iz} \right]_{-a}^{a} - \int_{-a}^{a} w'_\epsilon(x) \frac{e^{izx}}{iz}\, dx.$$

Here, $w_\epsilon(a) = e_n(a) = (-1)^n$ and $w_\epsilon(-a) = e_n(-a) = (-1)^n$. Moreover, $w'_\epsilon(x) = e'_n(x)(1 - \eta_\epsilon(x)) - e_n(x)\eta'_\epsilon(x)$. Since $|\eta'_\epsilon(x)| \ll \epsilon^{-1}$ on intervals of total length 2ϵ, we have $\int |\eta'_\epsilon(x)|\, dx = O(1)$. Hence

$$\|w'_\epsilon\|_{L^1} \le C_1\epsilon + O(1) \le C_2,$$

where C_2 is independent of ϵ. Consequently, there exists a constant $M > 0$ independent of ϵ such that

$$|\widehat{w_\epsilon}(z)| \le \frac{|w_\epsilon(a)| + |w_\epsilon(-a)| + \|w'_\epsilon\|_{L^1}}{|z|} \le \frac{M}{|z|}. \tag{3.4}$$

Finally, we show that $I_\epsilon \to 0$. We split the domain of integration into two parts at $|z| = 1/\epsilon$. For $|z| \le 1/\epsilon$, (3.3) gives

$$\int_{|z| \le 1/\epsilon} \log(3 + |z|) |\widehat{w_\epsilon}(z)|^2\, dz \le (2\epsilon)^2 \int_{-1/\epsilon}^{1/\epsilon} \log(3 + |z|)\, dz \ll \epsilon \log(1/\epsilon).$$

For $|z| > 1/\epsilon$, (3.4) gives

$$\int_{|z| > 1/\epsilon} \log(3 + |z|) |\widehat{w_\epsilon}(z)|^2\, dz \le 2 \int_{1/\epsilon}^{\infty} \log(3 + z) \frac{M^2}{z^2}\, dz \ll \epsilon \log(1/\epsilon).$$

Combining these estimates, we obtain

$$\|e_n - v_\epsilon\|_Q^2 = I_\epsilon + \|w_\epsilon\|_{L^2}^2 \ll \epsilon \log(1/\epsilon) \to 0 \qquad (\epsilon \to 0).$$

Therefore, $v_\epsilon \in C_c^\infty(-a, a)$ converges to e_n with respect to the form norm $\|\cdot\|_Q$. Hence every $e_n \in E$ belongs to $\overline{C_c^\infty(-a, a)}^{\|\cdot\|_Q}$ $(\subset \mathfrak{D}(\overline{Q}_B))$, and the inclusion (3.2) follows. $\square$

4. Proof of Theorem 1.3

4.1. **Compact embedding result.** The following proposition is the main analytic ingredient in the proof of Theorem 1.3.

Proposition 4.1. *For a bounded interval I, the embedding $H^{\log}(I) \hookrightarrow L^2(I)$ is compact. In other words, every bounded sequence in $H^{\log}(I)$ admits a subsequence converging in $L^2(I)$.*

Proof. The proof is essentially the same as that of [4, Theorem 3.6]. □

This compact embedding is also the key ingredient in the proof of the discreteness of the spectrum of A_a in [4, Theorem 3.6]. By [12, Proposition 10.6], it suffices to prove that the embedding $(\mathfrak{D}(Q_W^a), \|\cdot\|_{Q_W^a}) \hookrightarrow (L^2(-1,1), \|\cdot\|_{L^2})$ is compact. This follows readily from Proposition 4.1.

4.2. **Scaling transformation of the Rayleigh quotient.** Using (1.8), we transfer the Rayleigh quotient in (1.7) to the fixed interval $[-1,1]$ by a scaling transformation. From definition (1.6), the Rayleigh quotient associated with B_a for $v \in H_0^1(-a,a)$ is given by

$$\frac{\langle B_a v, v\rangle_{L^2}}{\|v\|_{L^2}^2} = \frac{\langle G_a Dv, Dv\rangle_{L^2}}{\|v\|_{L^2}^2} = \frac{\displaystyle\int_{-a}^{a}\int_{-a}^{a} g(x-y)v'(y)\overline{v'(x)}\,dxdy}{\displaystyle\int_{-a}^{a}|v(x)|^2 dx}.$$

Setting $w(t) = v(at)$ on the fixed interval $[-1,1]$, we define

$$q_a(w) := \int_{-1}^{1}\int_{-1}^{1} \frac{1}{a} g(a(x-y))w'(x)\overline{w'(y)}\,dx\,dy. \tag{4.1}$$

Then a change of variables gives

$$\frac{\langle B_a v, v\rangle_{L^2}}{\|v\|_{L^2}^2} = \frac{q_a(w)}{\|w\|_{L^2}^2}, \qquad w(t) = v(at) \in H_0^1(-1,1). \tag{4.2}$$

We write

$$R(a,w) := \frac{q_a(w)}{\|w\|_{L^2}^2}. \tag{4.3}$$

For the second term on the right-hand side of (2.3), we have

$$\int_{-a}^{a} |v(x)|^2 \log(a^2-x^2)\,dx = 2\|v\|_{L^2}^2 \log a + \int_{-1}^{1} |v(ax)|^2 \log(1-x^2)\,dx.$$

Thus, defining

$$\mathcal{L}(w) := \frac{1}{4}\int_{-1}^{1}\int_{-1}^{1} \frac{|w(x)-w(y)|^2}{|x-y|}\,dx\,dy - \frac{1}{2}\int_{-1}^{1} |w(x)|^2 \log(1-x^2)\,dx \tag{4.4}$$

for $v \in H_0^1(-1,1)$ (which should not be confused with the case $a = 1$ of (2.3)), we deduce from (2.5) that

$$\begin{aligned} R(a,w) = &-\log a - (2A+1) + \frac{\mathcal{L}(w)}{\|w\|_{L^2}^2} \\ &- \frac{1}{\|w\|_{L^2}^2} \sum_{n \le e^{2a}} \frac{\Lambda(n)}{\sqrt{n}} \left(\int_{-1}^{1-\frac{\log n}{a}} w\left(t + \frac{\log n}{a}\right)\overline{w(x)}\,dx \right. \\ &\qquad \left. + \int_{-1+\frac{\log n}{a}}^{1} w\left(x - \frac{\log n}{a}\right)\overline{w(x)}\,dx \right) \\ &- \frac{a}{\|w\|_{L^2}^2}\int_{-1}^{1}\int_{-1}^{1} r''(a(x-y))w(y)\overline{w(x)}\,dxdy \end{aligned} \tag{4.5}$$

for $w \in H_0^1(-1,1)$.

For the quadratic form $\mathcal{L}$ defined by (4.4) for $w \in \mathfrak{D}(\mathcal{L}) := H_0^1(-1,1)$, one can show in the same way as in the proof of (2.6) that

$$\mathcal{L}(w) = \frac{1}{2\pi}\int_{-\infty}^{\infty}\big(\log|z| + C_0\big)|\widehat{w}(z)|^2\,dz. \tag{4.6}$$

Hence the form norm $\|w\|_{\mathcal{L}}^2 := \mathcal{L}(w) + \|w\|_{L^2}^2$ is equivalent to $\int_{-\infty}^{\infty}(1+\log^+|z|)|\widehat{w}(z)|^2\,dz$. Therefore, the completion of $\mathfrak{D}(\mathcal{L})$ with respect to $\|\cdot\|_{\mathcal{L}}$ can be identified with a subspace of $H^{\log}(-1,1)$. In particular, formula (4.6) is valid not only for $w \in \mathfrak{D}(\mathcal{L})$, but also for $w \in \mathfrak{D}(\overline{\mathcal{L}})$.

To prove Theorem 1.3, it remains to establish the upper semicontinuity

$$\limsup_{a\to a_0}\lambda_a \le \lambda_{a_0}. \tag{4.7}$$

and lower semicontinuity

$$\liminf_{n\to\infty}\lambda_{a_n} \ge \lambda_{a_0}. \tag{4.8}$$

From (4.2) and Theorem 1.1, we obtain for the Rayleigh quotient associated with the closure $\bar{q}_a$ of q_a that

$$\frac{Q_W^a(v)}{\|v\|_{L^2}^2} = \frac{\bar{q}_a(w)}{\|w\|_{L^2}^2}, \quad v(at) = w(t) \in \mathfrak{D}(\bar{q}_a). \tag{4.9}$$

By (4.5), (4.6), and the definition of $H^{\log}(-1,1)$, we obtain $\mathfrak{D}(\bar{q}_a) \subset H^{\log}(-1,1)$ for every $a > 0$. Hence, by (1.7) and (4.9), it suffices to study the infimum of

$$\bar{R}(a,w) := \frac{\bar{q}_a(w)}{\|w\|_{L^2}^2}.$$

4.3. **Proof of upper semicontinuity.** Let q_a be the quadratic form defined in (4.1), and let $R(a,w)$ be the Rayleigh quotient defined in (4.3). We choose $w_0 \in \mathfrak{D}(\bar{q}_{a_0}) \subset H^{\log}(-1,1)$ such that $\lambda_{a_0} = \bar{R}(a_0,w_0)$, and normalize it by $\|w_0\|_{L^2} = 1$. To avoid referring to pointwise values of w_0 and $\widehat{w_0}$, we choose a sequence $w_n \in C_c^\infty(-1,1)$ such that $w_n \to w_0$ with respect to the form norm $\|\cdot\|_0$ (the form norm associated with q_{a_0}), and $\|w_n\|_{L^2} = 1$. Such a sequence exists by the proof of Theorem 1.1. Since $C_c^\infty(-1,1) \subset \mathfrak{D}(q_a)$, each w_n belongs to $\mathfrak{D}(q_a)$ for every a, and $R(a,w_n)$ can be expressed in terms of w_n and $\widehat{w_n}$.

Fix n. By the definition of the infimum $\lambda_a \le R(a,w_n)$ for every a. Moreover, (4.5) implies that $R(a,w_n) \to R(a_0,w_n)$ as $a \to a_0$. Since $\|w_n\|_{L^2} = 1$, we obtain

$$\limsup_{a\to a_0}\lambda_a \le R(a_0,w_n) = q_{a_0}(w_n).$$

Passing to the limit as $n \to \infty$, we obtain

$$\limsup_{a\to a_0}\lambda_a \le \lim_{n\to\infty} q_{a_0}(w_n) = \bar{q}_{a_0}(w_0) = \bar{R}(a_0,w_0) = \lambda_{a_0}.$$

Thus, (4.7) holds. □

4.4. **Proof of lower semicontinuity.** By (4.5), we decompose $\bar{q}_a = \bar{q}^0 + \bar{q}_a^1$, where $\bar{q}^0(w) = \overline{\mathcal{L}}(w) - (2A+1)\|w\|_{L^2}^2$ and $\bar{q}_a^1$ denotes the remaining terms in (4.5). Let (a_n) be a sequence with $a_n \to a_0$, and let $w_n \in \mathfrak{D}(\bar{q}_{a_n}) \subset H^{\log}(-1,1)$ be a sequence such that $\lambda_{a_n} = \bar{R}(a_n,w_n)$ and $\|w_n\|_{L^2} = 1$. We first show that there exists a subsequence of (w_n) which converges in L^2 to some limit w_*. Fix an arbitrary $w \in H_0^1(-1,1) \subset H^{\log}(-1,1)$. Then $\lambda_{a_n} \le R(a_n,w)$, and since $R(a,w)$ is continuous in a by the formula (4.5), the sequence (λ_{a_n}) is bounded above.

Since $\lambda_{a_n} = \bar{q}_{a_n}(w_n)$ by $\lambda_{a_n} = \bar{R}(a_n,w_n)$ and $\|w_n\|_{L^2} = 1$, the sequence $\bar{q}_{a_n}(w_n)$ is bounded above. As (a_n) remains in a compact neighborhood of a_0, the estimate $\bar{q}_a^1(v) = O(\|v\|_{L^2}^2)$ holds uniformly in a. It follows that the terms other than $\overline{\mathcal{L}}(w_n)$

in (4.5) are uniformly bounded. Consequently, $\overline{\mathcal{L}}(w_n)$ is bounded above. Since $\overline{\mathcal{L}}$ is bounded below by definition, the sequence $\overline{\mathcal{L}}(w_n)$ is bounded. Together with (4.6), this shows that (w_n) is bounded in $H^{\log}(-1,1)$. Proposition 4.1 therefore implies that (w_n) is relatively compact in L^2, and we may extract a subsequence converging in L^2. We relabel it again as (w_n) and denote its limit by w_*.

The explicit expression for $\bar{q}_a^1$ shows that it is a finite sum of quadratic forms associated with translation operators and integral operators with continuous kernels. Thus $\bar{q}_a^1(w)$ depends continuously on (a, w) near (a_0, w_*), and therefore

$$\lim_{n\to\infty} \bar{q}_{a_n}^1(w_n) = \bar{q}_{a_0}^1(w_*).$$

On the other hand, $\bar{q}^0$ is an a-independent lower bounded closed quadratic form on $L^2(-1,1)$. Hence, by [12, Proposition 10.1], it is lower semicontinuous with respect to L^2-convergence:

$$\liminf_{n\to\infty} \bar{q}^0(w_n) \geq \bar{q}^0(w_*).$$

Combining these estimates, we obtain

$$\liminf_{n\to\infty} \lambda_{a_n} = \liminf_{n\to\infty} \left(\bar{q}^0(w_n) + \bar{q}_{a_n}^1(w_n)\right) \geq \bar{q}^0(w_*) + \bar{q}_{a_0}^1(w_*) = \bar{q}_{a_0}(w_*).$$

Since $w_n \to w_*$ in $L^2(-1,1)$ and $\|w_n\|_{L^2} = 1$ for all n, we have $\|w_*\|_{L^2} = 1$. Therefore, $\bar{R}(a_0, w_*) = \bar{q}_{a_0}(w_*)$, and by the definition of λ_{a_0} as the infimum of $\bar{R}(a_0, \cdot)$, $\bar{q}_{a_0}(w_*) = \bar{R}(a_0, w_*) \geq \lambda_{a_0}$. Consequently,

$$\liminf_{n\to\infty} \lambda_{a_n} \geq \bar{q}_{a_0}(w_*) \geq \lambda_{a_0},$$

which proves (4.8). □

4.5. **A remark on the continuity of λ_a via parity decomposition.** Since the kernel $g(x-y)$ is even, G_a commutes with the parity operator $(Ju)(x) = u(-x)$, i.e. $G_a J = J G_a$. It follows that both $B_a = D^* G_a D$ and its Friedrichs extension A_a commute with J. In particular, each eigenspace $E(\lambda)$ of A_a admits a decomposition into even and odd subspaces: $E(\lambda) = E^+(\lambda) \oplus E^-(\lambda)$. For the lowest eigenvalue space $E(\lambda_a) = E^+(\lambda_a) \oplus E^-(\lambda_a)$, the assumption in [4, Theorem 5.10] amounts to requiring that $\dim E^+(\lambda_a) = 1$ and $E^-(\lambda_a) = \{0\}$.

In view of the parity decomposition and Corollary 1.2, we define

$$\lambda_a^\bullet := \inf_{0\neq v\in H_\bullet^1(-a,a)} \frac{\langle B_a v, v\rangle_{L^2}}{\|v\|_{L^2}^2}, \qquad \bullet \in \{+,-\}.$$

Then the global infimum λ_a in (1.7) can be expressed as

$$\lambda_a = \min(\lambda_a^+, \lambda_a^-). \tag{4.10}$$

Thus the continuity of λ_a follows from that of $\lambda_a^\pm$, since the minimum of two continuous functions is continuous. The continuity of $\lambda_a^\pm$ is asserted in [1, Theorem 5], although the details of the proof are not fully provided there.

We now verify that (4.10) holds. Every $v \in H_0^1(-a,a)$ admits a unique decomposition into even and odd components: $v = v^+ + v^-$. Since differentiation reverses parity and G_a preserves it, Dv^+ and $G_a(Dv^+)$ are odd, whereas Dv^- and $G_a(Dv^-)$ are even. Expanding the quadratic form in the numerator,

$$\langle G_a Dv, Dv\rangle = \langle G_a(Dv^+), Dv^+\rangle + \langle G_a(Dv^-), Dv^-\rangle,$$

since the cross terms vanish due to the orthogonality. Similarly, $\|v\|_{L^2}^2 = \|v^+\|_{L^2}^2 + \|v^-\|_{L^2}^2$ for the denominator. Hence the Rayleigh quotient on $H_0^1(-a,a)$ can be written as

$$\frac{\langle G_a Dv, Dv\rangle}{\|v\|_{L^2}^2} = \frac{\langle G_a(Dv^+), Dv^+\rangle + \langle G_a(Dv^-), Dv^-\rangle}{\|v^+\|^2 + \|v^-\|^2}.$$

For each component we have $\langle G_a(Dv^\bullet), Dv^\bullet\rangle \geq \lambda_a^\bullet \|v^\bullet\|^2$ for $\bullet \in \{+,-\}$ by definition. Therefore,

$$\frac{\langle G_a Dv, Dv\rangle}{\|v\|_{L^2}^2} \geq \frac{\lambda_a^+\|v^+\|^2 + \lambda_a^-\|v^-\|^2}{\|v^+\|^2 + \|v^-\|^2} \geq \min(\lambda_a^+, \lambda_a^-),$$

which yields $\lambda_a \geq \min(\lambda_a^+, \lambda_a^-)$. Conversely, restricting the Rayleigh quotient to $H_+^1(-a,a)$ and $H_-^1(-a,a)$ immediately yields $\lambda_a \leq \min(\lambda_a^+, \lambda_a^-)$. Thus, (4.10) is established.

5. Proof of Theorem 1.4

5.1. **Positivity of the lower bound.** By (4.9), the behavior of λ_a for small $a > 0$ is determined by the asymptotics of the Rayleigh quotient $R(a,w)$ in (4.3) as $a \to 0+$. From (4.5), we have

$$R(a,v) = \log\frac{1}{a} - (2A+1) + \frac{\mathcal{L}(v)}{\|v\|^2} + O(a) \tag{5.1}$$

for $0 < a < (1/2)\log 2$ and $v \in H_0^1(-1,1)$, so it suffices to study $\mathcal{L}(v)/\|v\|^2$. Note that this quantity is independent of a. In general, the infimum of $\mathcal{L}(v)/\|v\|^2$ need not coincide with that of $R(a,v)$, and hence their minimizers need not agree, but this distinction will not be relevant in what follows. By (4.6), the domain $\mathfrak{D}(\bar{\mathcal{L}})$ of the closure $\bar{\mathcal{L}}$ is contained in $H^{\log}(-1,1)$. Furthermore, the self-adjoint operator T associated with $\bar{\mathcal{L}}$ has discrete spectrum. This can be proved by the same argument as in the proof of [4, Theorem 3.6], using Proposition 4.1.

The term $\log(1/a)$ on the right-hand side of (5.1) diverges to $+\infty$ as $a \to 0+$. Since the closure $\bar{\mathcal{L}}$ is a lower bounded closed quadratic form, it is lower semicontinuous [12, Proposition 10.1]. We claim that $\inf_{\|v\|=1}\bar{\mathcal{L}}(v) > 0$. Indeed, it follows directly from the definition that $\mathcal{L}(v) \geq 0$. If there existed a sequence (v_n) such that $\mathcal{L}(v_n) \to 0$, then the first term in (4.4) would also tend to 0, hence a subsequence of (v_n) converges in $L^2(-1,1)$ to a constant function by (4.6) and Proposition 4.1. However, the second term in (4.4) is strictly positive for constant functions. Therefore, for sufficiently small $a > 0$, we obtain $\lambda_a = \inf_w R(a,w) > 0$.

5.2. **Positivity improving property.** The positivity improving property of the semigroup associated with $\bar{\mathcal{L}}$ established in this subsection will play a key role in proving the simplicity of the lowest eigenvalue. From (4.4), it is easy to see that $\mathcal{L}(|v|) \leq \mathcal{L}(v)$ holds for all $v \in \mathfrak{D}(\mathcal{L})$. Hence a minimizer v_0 of the Rayleigh quotient $\bar{\mathcal{L}}(v)/\|v\|^2$ may be chosen to be nonnegative.

To show that v_0 is positive, rather than merely nonnegative, we apply the theory of Dirichlet forms [5]. A symmetric closed quadratic form $\mathcal{E}$ with domain $\mathcal{F}$ is called a Dirichlet form if it satisfies the Markov property. That is, for every $v \in \mathcal{F}$, the function $v^\sharp := \phi\circ v = (0\vee v)\wedge 1$, $\phi(t) := \min(1, \max(0,t))$, belongs to $\mathcal{F}$ and satisfies $\mathcal{E}(v^\sharp) \leq \mathcal{E}(v)$. Note that ϕ is Lipschitz continuous.

We claim that $v^\sharp \in \mathfrak{D}(\bar{\mathcal{L}})$ and $\bar{\mathcal{L}}(v^\sharp) \leq \bar{\mathcal{L}}(v)$ for every $v \in \mathfrak{D}(\bar{\mathcal{L}})$. As the domain $\mathfrak{D}(\mathcal{L})$ is dense in $\mathfrak{D}(\bar{\mathcal{L}})$, it suffices to verify these properties on $\mathfrak{D}(\mathcal{L})$ using the Beurling–Deny representation (4.4) of $\bar{\mathcal{L}}$ [5, Theorem 3.2.1]. For the jumping part, the Lipschitz continuity of ϕ yields

$$|v^\sharp(x) - v^\sharp(y)|^2 = |\phi(v(x)) - \phi(v(y))|^2 \leq |v(x) - v(y)|^2.$$

Multiplying by the kernel $|x-y|^{-1}$ and integrating, we obtain

$$\frac{1}{2}\int_{-1}^{1}\int_{-1}^{1} \frac{|v^\sharp(x) - v^\sharp(y)|^2}{|x-y|}\,dx\,dy \leq \frac{1}{2}\int_{-1}^{1}\int_{-1}^{1} \frac{|v(x) - v(y)|^2}{|x-y|}\,dx\,dy.$$

For the killing part, $|v^\sharp(x)|^2 \le |v(x)|^2$ by $|\phi(a)| \le |a|$, and $\kappa(x) := -\log(1-x^2) \ge 0$. Hence

$$\int_{-1}^{1} \kappa(x)|v^\sharp(x)|^2\,dx \le \int_{-1}^{1} \kappa(x)|v(x)|^2\,dx.$$

Combining these estimates, we obtain $v^\sharp \in \mathfrak{D}(\bar{\mathcal{L}})$ and $\bar{\mathcal{L}}(v^\sharp) \le \bar{\mathcal{L}}(v)$. Therefore $(\bar{\mathcal{L}}, \mathfrak{D}(\bar{\mathcal{L}}))$ satisfies the Markov property and hence defines a Dirichlet form.

We next show that $(\bar{\mathcal{L}}, \mathfrak{D}(\bar{\mathcal{L}}))$ is irreducible. Recall that a Dirichlet form is called irreducible if every invariant set is trivial, that is, either it or its complement has measure zero. Let $A \subset (-1,1)$ be an $\bar{\mathcal{L}}$-invariant set. By the characterization of invariant sets in [9, p. 173], $\bar{\mathcal{L}}(1_A u, 1_{A^c} u) = 0$ for all $u \in \mathfrak{D}(\bar{\mathcal{L}})$, where $A^c = (-1,1) \setminus A$. Since $\mathfrak{D}(\mathcal{L})$ is a core of $\mathfrak{D}(\bar{\mathcal{L}})$, it suffices to test the condition on $\mathfrak{D}(\mathcal{L})$. We have

$$\mathcal{L}(1_A u, 1_{A^c} u) = -\frac{1}{2}\int_A \int_{A^c} \frac{u(x)u(y)}{|x-y|}\,dx\,dy = 0$$

for all $u \in \mathfrak{D}(\mathcal{L})$, since

$$\mathcal{L}(u,v) = \frac{1}{4}\int_{-1}^{1}\int_{-1}^{1} \frac{(u(x)-u(y))\overline{(v(x)-v(y))}}{|x-y|}\,dx\,dy - \frac{1}{2}\int_{-1}^{1} u(x)\overline{v(x)}\log(1-x^2)\,dx.$$

Because the jumping kernel $|x-y|^{-1}$ is positive on $(-1,1)^2$, the above identity cannot hold when both A and A^c have positive measure. Thus either A or A^c has measure zero, and $\bar{\mathcal{L}}$ is irreducible.

By [9, Theorem 1.4], the irreducibility implies that the associated semigroup $\{e^{-tT} \mid t > 0\}$ is positivity improving. In other words, for every nonnegative function $f \in L^2(-1,1)$ with $f \not\equiv 0$, one has $(e^{-tT} f)(x) > 0$ for a.e. $x \in (-1,1)$. In particular, the nonnegative eigenfunction v_0 of T is positive almost everywhere.

5.3. **Simplicity of the lowest eigenvalue and parity of eigenfunctions.** Suppose that the lowest eigenvalue of T admits two orthogonal eigenfunctions. Then, by the result of the previous subsection, eigenfunctions corresponding to the lowest eigenvalue may be chosen to be positive almost everywhere. However, two functions that are positive almost everywhere cannot be orthogonal to each other, which leads to a contradiction. Hence the eigenspace corresponding to the lowest eigenvalue of T is one-dimensional, and the lowest eigenvalue is simple.

Next, decompose the quadratic form $\bar{q}_a(v)$ according to (5.1). The term $(\log(1/a) - (2A+1))\|v\|_{L^2}^2$ merely shifts the spectrum, so the asymptotic behavior of λ_a is governed by the operator associated with $\bar{\mathcal{L}}(v) + O(a)\|v\|_{L^2}^2$. By [7, Chapter VIII, Section 3, Theorems 3.6 and 3.15], as $a \to 0+$, this operator has the same multiplicity for its lowest eigenvalue as T. Therefore, the lowest eigenvalue of the perturbed operator, and hence λ_a, is simple for sufficiently small $a > 0$.

As discussed in Section 4.5, the eigenspaces of the self-adjoint operator associated with $\bar{q}_a$ decompose into even and odd parts. In particular, since the lowest eigenvalue is simple, its eigenfunction must be either even or odd. However, since the eigenfunction corresponding to the lowest eigenvalue has a fixed sign, it must be even.

6. Proof of Theorem 1.5

6.1. **Symmetry of the minimal operator.** Choose $\lambda < \lambda_a$, and let $\mathcal{H}(T_a)$ denote the Hilbert space obtained as the completion of $C_c^\infty(-a,a)$ with respect to the norm $\|\cdot\|_{T_a}$ defined by (1.9). Define

$$\langle v_1, v_2\rangle_{T_a} := \langle T_a v_1, v_2\rangle_{L^2}$$

so that $\|v\|_{T_a}^2 = \langle v, v\rangle_{T_a}$. Let $\mathscr{D}_a$ be the operator on $\mathcal{H}(T_a)$ with domain (1.10) acting as $\mathscr{D}_a = i\,d/dx$. Since the canonical map $C_c^\infty(-a,a) \to \mathcal{H}(T_a)$ is injective and the image is dense, the operator $\mathscr{D}_a$ is densely defined.

Lemma 6.1. *The operator $\mathscr{D}_a$ is symmetric on $\mathcal{H}(T_a)$.*

Proof. We show that $\langle \mathscr{D}_a u, v\rangle_{T_a} = \langle u, \mathscr{D}_a v\rangle_{T_a}$ for all $u, v \in \mathfrak{D}(\mathscr{D}_a)$. By the definition of the inner product on $\mathcal{H}(T_a)$ and the self-adjointness of A_a, we have

$$\langle \mathscr{D}_a u, v\rangle_{T_a} = \langle A_a(iu'), v\rangle_{L^2} - \lambda\langle iu', v\rangle_{L^2} = \langle iu', A_a v\rangle_{L^2} - \lambda\langle iu', v\rangle_{L^2}.$$

According to the distribution kernel representation (2.10), the function $A_a v$ is the convolution of v with the tempered distribution $k := -g''$. Hence, by the general theory of tempered distributions, $A_a v \in C^\infty(\mathbb{R})$ whenever $v \in \mathfrak{D}(\mathscr{D}_a)$. Using this fact together with the boundary conditions $u(a) = u(-a) = 0$, integration by parts gives $\langle iu', A_a v\rangle_{L^2} = \langle u, i(A_a v)'\rangle_{L^2}$, and $\langle iu', v\rangle_{L^2} = \langle u, iv'\rangle_{L^2}$. Furthermore, using (2.10) and the boundary conditions $v(a) = v(-a) = 0$, we obtain by integration by parts that

$$(A_a v)'(x) = \frac{d}{dx}\int_{-a}^{a} k(x-y)v(y)\,dy = -\int_{-a}^{a}\frac{d}{dy}k(x-y)v(y)\,dy = \int_{-a}^{a} k(x-y)v'(y)\,dy.$$

Therefore, $\langle u, i(A_a v)'\rangle_{L^2} = \langle u, A_a(iv')\rangle_{L^2}$, and hence

$$\langle \mathscr{D}_a u, v\rangle_{T_a} = \langle u, A_a(iv')\rangle_{L^2} - \lambda\langle u, iv'\rangle_{L^2} = \langle A_a u, iv'\rangle_{L^2} - \lambda\langle u, iv'\rangle_{L^2} = \langle u, \mathscr{D}_a v\rangle_{T_a}.$$

This proves the desired identity. □

6.2. **Adjoint of the minimal operator.** We now derive the adjoint operator $\mathscr{D}_a^*$ of $\mathscr{D}_a$ on $\mathcal{H}(T_a)$. For a vector $v \in \mathcal{H}(T_a)$ to belong to the domain $\mathfrak{D}(\mathscr{D}_a^*)$ of $\mathscr{D}_a^*$, the linear functional $u \mapsto \langle \mathscr{D}_a u, v\rangle_{T_a}$ must be continuous on $\mathfrak{D}(\mathscr{D}_a)$ with respect to the norm of $\mathcal{H}(T_a)$. Since $\mathscr{D}_a$ is symmetric, $\mathfrak{D}(\mathscr{D}_a) \subset \mathfrak{D}(\mathscr{D}_a^*)$. Moreover, by the calculation in the previous section, we have $\langle \mathscr{D}_a u, v\rangle_{T_a} = \langle u, i(A_a v)'\rangle_{L^2} - \lambda\langle u, iv'\rangle_{L^2}$ for $u \in \mathfrak{D}(\mathscr{D}_a)$. If this functional is continuous with respect to $\|\cdot\|_{T_a}$, then it is in particular continuous with respect to the L^2-norm. Hence it is necessary that $v' \in L^2(-a,a)$ and $(A_a v)' \in L^2(-a,a)$. Therefore,

$$\mathfrak{D}(\mathscr{D}_a^*) \;\subset\; \{\, v \in \mathcal{H}(T_a) \mid v \in H^1(-a,a),\ A_a v \in H^1(-a,a)\,\}.$$

The condition that $v \in \mathfrak{D}(\mathscr{D}_a^*)$ and $\mathscr{D}_a^* v = g \in \mathcal{H}(T_a)$ means that $\langle \mathscr{D}_a u, v\rangle_{T_a} = \langle u, g\rangle_{T_a}$ holds for every $u \in \mathfrak{D}(\mathscr{D}_a)$. By the calculation in the previous section, we have

$$\langle \mathscr{D}_a u, v\rangle_{T_a} = \langle u, i(T_a v)'\rangle_{L^2}$$

for $u \in \mathfrak{D}(\mathscr{D}_a)$. Since $\mathfrak{D}(\mathscr{D}_a)$ is dense in $\mathcal{H}(T_a)$, it follows that

$$T_a(\mathscr{D}_a^* v) = i(T_a v)'.$$

If $v \in \mathfrak{D}(\mathscr{D}_a)$, then this identity becomes

$$T_a(\mathscr{D}_a^* v) = i(T_a v)' = T_a(iv') = T_a(\mathscr{D}_a v),$$

and hence $\mathscr{D}_a^*$ agrees with $\mathscr{D}_a$. The eigenvalue equations $\mathscr{D}_a^* v = \pm i\, v$ are equivalent to $i(T_a v)' = \pm i(T_a v)$. Since $\lambda < \lambda_a$, the operator T_a is invertible on $L^2(-a,a)$. Hence the equations

$$(T_a v)(x) = C_+ e^{x}, \qquad (T_a v)(x) = C_- e^{-x},$$

admit unique solutions, and these functions span the eigenspaces corresponding to the eigenvalues $\pm i$. Therefore we obtain the following result.

Lemma 6.2. *$\mathscr{D}_a$ has the deficiency indices $(1,1)$.*

6.3. **Eigenfunctions of $\mathscr{D}_a^*$.** Let f_z be an eigenfunction of $\mathscr{D}_a^*$ corresponding to an eigenvalue $z \in \mathbb{C}$. The equation $\mathscr{D}_a^* f_z = z f_z$ implies that $(T_a f_z)(x) = C\exp(-izx)$, which uniquely determines f_z for each fixed $C \neq 0$, since T_a is invertible. Hence the eigenspace corresponding to the eigenvalue z is at most one-dimensional. We now give a concrete characterization of this eigenspace.

Let $z \in \mathbb{C}$. Since the Fourier transform is continuous on $L^2(-a,a)$ and $\|v\|_{T_a} \gg \|v\|_{L^2}$, the map $v \mapsto \widehat{v}(z)$ extends continuously to $\mathcal{H}(T_a)$. Hence, by the Riesz representation theorem, there exists a unique element $v_z \in \mathcal{H}(T_a)$ such that $\langle v, v_z\rangle_{T_a} = \widehat{v}(\bar{z})$ for all $v \in \mathcal{H}(T_a)$. For every $v \in \mathfrak{D}(\mathscr{D}_a)$,

$$\langle v, \mathscr{D}_a^* v_z\rangle_{T_a} = \langle \mathscr{D}_a v, v_z\rangle_{T_a} = \langle iv', v_z\rangle_{T_a} = \bar{z}\widehat{v}(\bar{z}) = \bar{z}\langle v, v_z\rangle_{T_a} = \langle v, z v_z\rangle_{T_a}.$$

Since $\mathfrak{D}(\mathscr{D}_a)$ is dense in $\mathcal{H}(T_a)$, it follows that $v_z \in \mathfrak{D}(\mathscr{D}_a^*)$ and $\mathscr{D}_a^* v_z = z v_z$. As $v_z \neq 0$, we conclude that the eigenspace of $\mathscr{D}_a^*$ corresponding to the eigenvalue z is one-dimensional and is spanned by v_z.

Let $e_z(x) := \exp(-izx)$. Since $e_z \in L^2(-a,a)$ and T_a is invertible, the equation $T_a w_z = e_z$ has a unique solution $w_z \in \mathfrak{D}(T_a)$. Moreover, for every $v \in \mathfrak{D}(\mathscr{D}_a)$, $\widehat{v}(\bar{z}) = \langle v, e_z\rangle_{L^2} = \langle v, T_a w_z\rangle_{L^2} = \langle v, w_z\rangle_{T_a}$. By the uniqueness in the Riesz representation theorem, we conclude that $w_z = v_z$. Hence $v_z \in \mathfrak{D}(T_a)$.

6.4. **Self-adjoint extensions of the minimal operator.** We compute the self-adjoint extensions of $\mathscr{D}_a$ using von Neumann's theory of deficiency indices ([11, Section X.1], [12, Section 3.2]). By the results of the previous subsection, the deficiency spaces $\mathrm{Ker}\,(\mathscr{D}_a^* \mp i)$ are one-dimensional and are spanned by vectors $v_\pm = C_\pm v_{\pm i}$ with the normalization $\|v_+\|_{T_a} = \|v_-\|_{T_a}$. Thus, every self-adjoint extension $\overline{\mathscr{D}}_{a,\theta}$ parametrized by $\theta \in [0, 2\pi)$ has domain

$$\begin{aligned}\mathfrak{D}(\overline{\mathscr{D}}_{a,\theta}) &= \{v_0 + c(v_+ + e^{i\theta}v_-) \mid v_0 \in \mathfrak{D}(\overline{\mathscr{D}}_a), c \in \mathbb{C}\}\\ &\subset \mathfrak{D}(\mathscr{D}_a^*) = \{v_0 + av_+ + bv_- \mid v_0 \in \mathfrak{D}(\overline{\mathscr{D}}_a), a, b \in \mathbb{C}\}.\end{aligned}$$

These domains are characterized by the boundary form

$$W(u,v) = \langle \mathscr{D}_a^* u, v\rangle_{T_a} - \langle u, \mathscr{D}_a^* v\rangle_{T_a}, \tag{6.1}$$

namely, $v \in \mathfrak{D}(\mathscr{D}_a^*)$ belongs to $\mathfrak{D}(\overline{\mathscr{D}}_{a,\theta})$ if and only if $W(v, w_\theta) = 0$ for $w_\theta = v_+ + e^{i\theta}v_-$. The action of $\overline{\mathscr{D}}_{a,\theta}$ is given by

$$\overline{\mathscr{D}}_{a,\theta}\Big(v_0 + c(v_+ + e^{i\theta}v_-)\Big) := \overline{\mathscr{D}}_a v_0 + ic(v_+ - e^{i\theta}v_-),$$

where $\overline{\mathscr{D}}_a$ is the closure of $\mathscr{D}_a$.

6.5. **Proof of Theorem 1.5.** We choose the basis vectors of the deficiency spaces so that $\mathscr{D}_a^* v_\pm = \pm i v_\pm$ and $\|v_+\|_{T_a} = \|v_-\|_{T_a}$. On the other hand, applying $\mathscr{D}_a^*$ to $w_\theta = v_+ + e^{i\theta}v_-$ yields

$$\mathscr{D}_a^* w_\theta = iv_+ - ie^{i\theta}v_-.$$

Substituting these equalities into the boundary form W, we obtain

$$W(v_z, w_\theta) = \langle z v_z, v_+ + e^{i\theta}v_-\rangle_{T_a} - \langle v_z, iv_+ - ie^{i\theta}v_-\rangle_{T_a}.$$

Expanding the right-hand side by linearity of the inner product and collecting the terms involving the inner products with v_+ and v_-, we obtain

$$W(v_z, w_\theta) = (z+i)\langle v_z, v_+\rangle_{T_a} + (z-i)e^{-i\theta}\langle v_z, v_-\rangle_{T_a} = 0. \tag{6.2}$$

Using the definition of the inner product on $\mathcal{H}(T_a)$ and the identity $\langle v, v_z\rangle_{T_a} = \widehat{v}(\bar{z})$, we obtain

$$\langle v_z, v_\pm\rangle_{T_a} = \overline{\langle v_\pm, v_z\rangle_{T_a}} = \overline{\widehat{v_\pm}(\bar{z})} = \int_{-a}^{a} \overline{v_\pm(x)}e^{-izx}\,dx. \tag{6.3}$$

Because $v_\pm \in \mathcal{H}(T_a) \subset L^2(-a,a) \subset L^1(-a,a)$, the above Fourier integrals define entire functions of z. Consequently, upon rewriting the orthogonality condition (6.2), we find that the eigenvalue z must satisfy the equation $W(a,\theta;z)=0$, where $W(a,\theta;z)$ is defined in (1.11). This proves the first assertion of the theorem.

Next, we show that all zeros of $W(a,\theta;z)$ are real. Suppose that a complex number $\lambda_0 \in \mathbb{C}\setminus\mathbb{R}$ satisfies $W(a,\theta;\lambda_0)=0$. By the characterization established above, this means that the corresponding v_{λ_0} is orthogonal to the deficiency vector w_θ with respect to the boundary form, that is, $W(v_{\lambda_0},w_\theta)=0$. By definition, every element satisfying this condition belongs to $\mathfrak{D}(\overline{\mathscr{D}}_{a,\theta})$. Consequently, v_{λ_0} is a nonzero element of the domain of the self-adjoint extension $\overline{\mathscr{D}}_{a,\theta}$ and satisfies $\overline{\mathscr{D}}_{a,\theta}v_{\lambda_0}=\lambda_0 v_{\lambda_0}$. Hence v_{λ_0} is a genuine eigenvector of the self-adjoint operator $\overline{\mathscr{D}}_{a,\theta}$ with eigenvalue λ_0. However, λ_0 must be real, since every eigenvalue of a self-adjoint operator is real. This contradiction shows that any complex number λ satisfying $W(a,\theta;\lambda)=0$ must lie on the real axis. □

7. Heuristic justification of the formula in Corollary 1.6

In this section, we assume RH. By Weil's positivity criterion, we have $Q_W \geq 0$ and hence $A_a > 0$ for all $a>0$. Under this assumption, for each $a>0$ we may take $\lambda=0$ in the definition (1.9) of $T_a = T_{a,\lambda}$. Hence we work with A_a in place of T_a, and consider the Hilbert spaces $\mathcal{H}(A_a) := \mathcal{H}(T_{a,0})$.

7.1. The Hilbert Space $\mathcal{H}(A_\infty)$. In order to study the behavior of the differential operator $\mathscr{D}_a$ on the Hilbert space $\mathcal{H}(A_a)$ as $a\to\infty$, we first introduce the Hilbert space $\mathcal{H}(A_\infty)$ and the corresponding differential operator $\mathscr{D}_\infty$. Here A_∞ is only a formal symbol and does not denote an actual operator.

Let $\mathfrak{D}(Q_W)$ denote the completion of $C_c^\infty(\mathbb{R})$ with respect to the form norm

$$\|v\|_{Q_W}^2 := Q_W(v) + \|v\|_{L^2}^2.$$

This space is a Hilbert space with respect to the form norm. $\mathcal{N} := \{v\in\mathfrak{D}(Q_W) \mid Q_W(v)=0\}$ is a closed subspace of $\mathfrak{D}(Q_W)$ with respect to the form norm. We then consider the quotient space

$$\mathcal{H}(A_\infty) := \mathfrak{D}(Q_W)/\mathcal{N}.$$

For $v\in\mathfrak{D}(Q_W)$, we denote its equivalence class by $[v]\in\mathcal{H}(A_\infty)$. We define the norm of $[v]$ by

$$\|[v]\|_{A_\infty}^2 := Q_W(v).$$

The Hilbert space $\mathcal{H}(A_\infty)$ can be identified with the completion $\mathcal{H}_W$ of $C_c^\infty(\mathbb{R})$ with respect to $Q_W(v)$ studied in [14].

Since $Q_W(v)\neq 0$ for any nonzero $v\in C_c^\infty(\mathbb{R})$, the map $v\mapsto[v]$ embeds $C_c^\infty(\mathbb{R})$ injectively into $\mathcal{H}(A_\infty)$, and its image is dense. Hence, by fixing representatives on $C_c^\infty(\mathbb{R})$, we define a densely defined operator $\mathscr{D}_\infty$ on $\mathcal{H}(A_\infty)$ by $\mathscr{D}_\infty[v] := [\mathscr{D}_\infty v] = [iv']$. This is well-defined since $[u]=[v]$ for $u,v\in C_c^\infty(\mathbb{R})$ implies $u=v$.

We now verify that $\mathscr{D}_\infty$ is symmetric on $\mathcal{H}(A_\infty)$ and hence closable. If $Q_W(w)=0$, it follows that $Q_W(w,v)=0$ for all v by the assumption $Q_W\geq 0$ and the Cauchy-Schwarz inequality $|Q_W(u,v)|^2 \leq Q_W(u)Q_W(v)$. This implies $Q_W(u+w,v)=Q_W(u,v)$ and $Q_W(u,v+w)=Q_W(u,v)$. Hence the sesquilinear form

$$Q_W([u],[v]) = Q_W(u,v) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}(-g''(x-y))\,u(y)\,\overline{v(x)}\,dxdy \tag{7.1}$$

is well-defined (cf. (2.9)). Moreover, the translation operator τ_t ($t\in\mathbb{R}$) defined by

$$\tau_t[u] := [\tau_t u], \quad (\tau_t u)(x) = u(x-t)$$

is well-defined, since $\tau_t[u+w] = [\tau_t(u+w)] = [\tau_t u]+[\tau_t w]$ and $Q_W([\tau_t w]) = Q_W(\tau_t w) = Q_W(w) = 0$ if $Q_W(w)=0$. Furthermore, by (7.1), the norm in $\mathcal{H}(A_\infty)$ is invariant under translations:

$$\|[v]\|_{A_\infty} = \|\tau_t[v]\|_{A_\infty}. \tag{7.2}$$

Since $\mathscr{D}_\infty$ commutes with convolution, it follows from (7.1) that

$$Q_W(\mathscr{D}_\infty[u],[v]) = Q_W([u],\mathscr{D}_\infty[v])$$

for $u, v \in C_c^\infty(\mathbb{R})$. Hence $\mathscr{D}_\infty$ is symmetric with respect to $\langle\cdot,\cdot\rangle_{A_\infty}$. Since every symmetric operator is closable, its closure will be denoted by $\bar{\mathscr{D}}_\infty$. Thus, for $[v] \in \mathfrak{D}(\bar{\mathscr{D}}_\infty)$, one can choose a sequence $v_n \in C_c^\infty(\mathbb{R})$ such that $[v_n] \to [v]$ and $\mathscr{D}_\infty[v_n] = [iv_n'] \to \bar{\mathscr{D}}_\infty[v]$ in $\mathcal{H}(A_\infty)$.

7.2. **Isomorphism with a de Branges space.** We now recall the isometric isomorphism between $\mathcal{H}(A_\infty)$ (denoted by $\mathcal{H}_W$ in [14]) and a certain de Branges space $\mathcal{B}$ established in [14]. Here it is sufficient to understand $\mathcal{B}$ as a reproducing kernel Hilbert space consisting of entire functions.

Let M be the operator of multiplication by an independent variable on $\mathcal{B}$ defined by $\mathfrak{D}(M) = \{F \in \mathcal{B} \mid zF(z) \in \mathcal{B}\}$ and $(MF)(z) = zF(z)$. The operator M is a densely defined closed symmetric operator with deficiency indices $(1,1)$, and it admits a one-parameter family of self-adjoint extensions parametrized by $\psi \in [0,\pi)$. In particular, the self-adjoint extension $M_{\pi/2}$ has purely discrete spectrum Γ, the set of zeros of $\xi(1/2-iz)$ in Section 3, and one can choose an orthonormal basis of $\mathcal{B}$ consisting of the corresponding eigenfunctions F_γ for $\gamma \in \Gamma$ [14, Section 6]. In this setting, defining $U : \mathcal{H}(A_\infty) \to \mathcal{B}$ by

$$U([v]) := \sum_{\gamma\in\Gamma} \sqrt{m_\gamma}\,\widehat{v}(\gamma)F_\gamma \tag{7.3}$$

yields an isometric isomorphism of Hilbert spaces [14, Theorems 1.1 and 1.4]:

$$(Q_W(v) =)\ \|[v]\|_{A_\infty}^2 = \|U([v])\|_{\mathcal{B}}^2. \tag{7.4}$$

This U is precisely the operator denoted by $\pi^{-1/2}\widehat{\mathcal{P}_D} = \pi^{-1/2}\widehat{\mathcal{P}}D$ in [14].

7.3. **Correspondence between $\mathscr{D}_\infty$ and M.** Via the isomorphism $U : \mathcal{H}(A_\infty) \to \mathcal{B}$, the operator M induces a self-adjoint operator on $\mathcal{H}(A_\infty)$. However, its relation to the differential operator $\bar{\mathscr{D}}_\infty$ is not immediately clear. On the other hand, the relation between the differential operator $\bar{\mathscr{D}}_\infty$ and the self-adjoint extension $M_{\pi/2}$ of M can be described in a simple way as follows.

If $v \in C_c^\infty(\mathbb{R})$, then $\widehat{v}$ is rapidly decreasing. Hence $[v]$ belongs to the domain of $M_{\pi/2}$. For each $\gamma \in \Gamma$, F_γ is an eigenfunction of $M_{\pi/2}$ with eigenvalue γ, and therefore

$$M_{\pi/2}U([v]) = \sum_\gamma \gamma\widehat{v}(\gamma)F_\gamma = \sum_\gamma \widehat{(iv')}(\gamma)F_\gamma = U([iv']) = U(\mathscr{D}_\infty[v]).$$

Let $\mathsf{D}_{\pi/2} := U^{-1}M_{\pi/2}U$. Then the above computation shows that $\mathsf{D}_{\pi/2}[v] = [iv'] = \mathscr{D}_\infty[v]$ for $v \in C_c^\infty(\mathbb{R})$. Since $\mathsf{D}_{\pi/2}$ is closed and extends $\mathscr{D}_\infty$, it follows that

$$\mathsf{D}_{\pi/2}[v] = \bar{\mathscr{D}}_\infty[v]$$

for all $[v] \in \mathfrak{D}(\bar{\mathscr{D}}_\infty)$. Thus, $\mathsf{D}_{\pi/2}$ is a self-adjoint extension of $\bar{\mathscr{D}}_\infty$.

Remark. More generally, for any self-adjoint extension M_ψ of M, one may define $\mathsf{D}_\psi := U^{-1}M_\psi U$. Since M_ψ is unitarily equivalent to D_ψ, the operator D_ψ is self-adjoint. It is natural to ask whether D_ψ can be characterized directly as a self-adjoint extension of $\bar{\mathscr{D}}_\infty$. We do not pursue this question here.

7.4. **An operator unitarily equivalent to M.** We now consider the operator on $\mathcal{H}(A_\infty)$ induced by M through U. More precisely, we define

$$\mathsf{D} := U^{-1}MU.$$

By construction, D and M are unitarily equivalent via U, and have the same deficiency indices. In particular, the deficiency indices of D are $(1,1)$.

Writing the Fourier transform as $\mathsf{F}v = \widehat{v}$, there exists a subspace $V(0) \subset L^2(0,\infty)$ such that the maps $\mathcal{B} \ni F \mapsto v_F := \mathscr{F}^{-1}F \in V(0)$ and $V(0) \ni v \mapsto [v] \in \mathcal{H}(A_\infty)$ are isomorphisms [14, Theorem 5.5]. In this setting, the operator on $V(0)$ unitarily equivalent to M via $\mathscr{F}^{-1} : \mathcal{B} \to V(0)$ is denoted by $D = \mathscr{F}^{-1}M\mathscr{F}$. Then, via the isomorphism $\iota : V(0) \to \mathcal{H}(A_\infty)$, the operator corresponding to D is given by $\mathsf{D} = \iota D \iota^{-1}$. In particular, we have $U = \mathscr{F}\iota^{-1}$. If $F \in \mathfrak{D}(M)$ and $F = \mathscr{F}v$, then

$$Dv = \mathscr{F}^{-1}(MF) = \mathscr{F}^{-1}(zF) = iv'.$$

Moreover, for $v \in V(0)$,

$$\mathsf{D}[v] = \iota(Dv) = \iota(iv') = [iv'].$$

We claim that for a given function u, the decomposition $u = v + w$ with $v \in V(0)$ and $[w] = 0$ is uniquely determined. Once this is established, we may define $\mathsf{D}[u] := \mathsf{D}[v]$ by writing $u = v + w$ with $v \in V(0)$ for $[u] \in \mathfrak{D}(\mathsf{D})$ (defined via the unitary correspondence with $\mathfrak{D}(M)$).

The uniqueness of v in the above decomposition is as follows. If $u = v + w$ and $u = \tilde{v} + \tilde{w}$ are two such decompositions, then $v - \tilde{v} = \tilde{w} - w$. Hence, if $v_0 := v - \tilde{v} \neq 0$, then $0 \neq v_0 \in V(0)$ and $[v_0] = 0$. It follows from (7.3) and (7.4) that $\widehat{v_0}(\gamma) = 0$ for all γ. Since $\mathscr{F} : V(0) \to \mathcal{B}$ is an isomorphism, we obtain $v_0 = 0$, a contradiction.

With this definition, the relation between

$$C_c^\infty(\mathbb{R}) \quad \text{and} \quad \mathfrak{D}(\mathsf{D})$$

is not clear. Indeed, for $u \in C_c^\infty(\mathbb{R})$ decomposed as $u = v + w$ with $v \in V(0)$ and $[w] = 0$, it is not clear whether $\mathscr{F}v \in \mathfrak{D}(M)$ holds.

7.5. **Infinitesimal generator of translations ($a = \infty$).** The previously defined family $(\tau_t)_{t\in\mathbb{R}}$ forms a strongly continuous one-parameter unitary group. Indeed, $\tau_t\tau_s = \tau_{t+s}$ is clear from the definition, and each τ_t is a unitary operator by (7.2). It remains to verify strong continuity, namely $\lim_{t\to 0} \|\tau_t[v] - [v]\|_{A_\infty} = 0$. To this end, using the isometric property of U, we estimate

$$\|U(\tau_t[v] - [v])\|_{\mathcal{B}}^2 = \sum_{\gamma\in\Gamma} m_\gamma |e^{-it\gamma} - 1|^2 |\widehat{v}(\gamma)|^2.$$

Since $A_\infty > 0$ implies RH, all elements of Γ are real. Since $|e^{-it\gamma} - 1|^2 \leq 4$, the dominated convergence theorem implies that the right-hand side tends to 0 as $t \to 0$. Hence τ_t is strongly continuous with respect to the norm of $\mathcal{H}(A_\infty)$.

Therefore, there exists a self-adjoint operator $\overline{\mathscr{D}}$ on $\mathcal{H}(A_\infty)$ such that

$$\tau_t = e^{it\overline{\mathscr{D}}} \quad (t \in \mathbb{R})$$

by Stone's theorem, where

$$\mathfrak{D}(\overline{\mathscr{D}}) = \left\{ [v] \in \mathcal{H}_\infty \;\middle|\; \lim_{\epsilon\to 0} \frac{\tau_\epsilon[v] - [v]}{\epsilon} \text{ exists} \right\}, \quad \overline{\mathscr{D}}[v] = -i \lim_{\epsilon\to 0} \frac{\tau_\epsilon[v] - [v]}{\epsilon}.$$

For $[v] \in \mathfrak{D}(\mathsf{D})$, we have

$$\overline{\mathscr{D}}[v] = [iv'] = \mathsf{D}[v]$$

by considering the action on the $\mathcal{B}$ side. Thus, the self-adjoint operator $\overline{\mathscr{D}}$ extends the closed symmetric operator D, whose deficiency indices are $(1,1)$. Moreover,

$$U(\overline{\mathscr{D}}[v]) = M_{\pi/2}U([v])$$

holds, that is, $\overline{\mathscr{D}} = \mathsf{D}_{\pi/2}$. Indeed,

$$\begin{aligned}(U\overline{\mathscr{D}}[v])(z) &= -i\lim_{\epsilon\to 0}\frac{(U\tau_\epsilon[v])(z)-(U[v])(z)}{\epsilon}\\ &= -i\lim_{\epsilon\to 0}\sum_{\gamma\in\Gamma}\sqrt{m_\gamma}\,\frac{e^{i\gamma\epsilon}-1}{\epsilon}\,\widehat{v}(\gamma)F_\gamma\\ &= \sum_{\gamma\in\Gamma}\sqrt{m_\gamma}\,\gamma\widehat{v}(\gamma)F_\gamma = (M_\psi U[v])(z).\end{aligned}$$

Viewed on $\mathcal{H}(A_\infty)$, $\overline{\mathscr{D}}$ is characterized as the unique self-adjoint extension of D that commutes with the (usual) translation. Moreover, the equality $\overline{\mathscr{D}} = \mathsf{D}_{\pi/2}$ asserts that a Hilbert–Pólya operator is realized as the infinitesimal generator of the translation group under the assumption $A_\infty > 0$ (RH).

Since $\overline{\mathscr{D}}$ acts as $v \mapsto iv'$ on $C_c^\infty(-a,a)$, to approximate the behavior as $a\to\infty$, it is natural to consider the minimal operator $\mathscr{D}_a$ for each $a>0$. Considering the self-adjoint extensions $\mathscr{D}_{a,\theta}$ of $\mathscr{D}_a$, it is expected that by choosing $\theta=\theta(a)$ appropriately,

$$\overline{\mathscr{D}}_{a,\theta} \ \to\ \overline{\mathscr{D}} \quad (a\to\infty)$$

in the sense of strong resolvent convergence.

7.6. **Operator picture via embedding.** We denote by $\mathcal{H}(A_a)$ the completion of $C_c^\infty(-a,a)$ with respect to $(Q_W|_{L^2(-a,a)})(v) = Q_W^a(v) = \langle A_a v, v\rangle_{L^2}$. Since $U(v)\neq 0$ for every nonzero $v\in C_c^\infty(-a,a)$, the space $C_c^\infty(-a,a)$ embeds injectively into $\mathcal{H}(A_\infty)$ via $v\mapsto [v]$ as in the case of $C_c^\infty(\mathbb{R})$. This embedding extends to an injective map

$$\mathcal{H}(A_a) \ \hookrightarrow\ \mathcal{H}(A_\infty).$$

Through this embedding, we may regard $\mathscr{D}_a$ and $\overline{\mathscr{D}}_{a,\theta}$ as operators on $\mathcal{H}(A_\infty)$, although they are no longer densely defined in $\mathcal{H}(A_\infty)$. However, we still have

$$\mathscr{D}_a = \mathscr{D}_\infty|_{\mathcal{H}(A_a)}$$

$$\bar{\mathscr{D}}_\infty \ \subset\ \mathsf{D}_{\pi/2} = \overline{\mathscr{D}}$$

and thus, as $a\to\infty$, it is natural to expect that $\overline{\mathscr{D}}_{a,\theta}$ approximates the self-adjoint extension $\mathsf{D}_{\pi/2} = \overline{\mathscr{D}}$ of $\bar{\mathscr{D}}_\infty$. Since the spectrum of $\mathsf{D}_{\pi/2}$ is given by Γ, one further expects that the eigenvalues of $\overline{\mathscr{D}}_{a,\theta}$ approximate Γ as $a\to\infty$.

7.7. **Factorization of A_∞.** By (7.4),

$$Q_W(v) = \langle A_\infty v, v\rangle_{L^2} = \langle Uv, Uv\rangle_{L^2} = \langle U^*Uv, v\rangle_{L^2}.$$

Using the notation of [14] and the factorization $U=\pi^{-1/2}\widehat{\mathcal{P}}D$, we obtain

$$A_\infty = U^*U = \pi^{-1}D^*\widehat{\mathcal{P}}^*\widehat{\mathcal{P}}D,\quad G=\pi^{-1}\widehat{\mathcal{P}}^*\widehat{\mathcal{P}}$$

under RH. Conversely, if one could show unconditionally that the kernel of U^*U coincides with $-g''(x-y)$, or equivalently that the kernel of $\pi^{-1}\widehat{\mathcal{P}}^*\widehat{\mathcal{P}}$ coincides with the kernel $g(x-y)-g(x)-g(-y)+g(0)$ of G, then it would follow that $A_\infty>0$, and hence RH would follow immediately. More precisely, using the notation of [14], one has unconditionally

$$(Uv)(z) = \pi^{-1/2}\int_{-\infty}^{\infty}\overline{\mathfrak{S}_x(\bar{z})}\,v'(x)\,dx,$$

and therefore RH would follow if one could establish unconditionally that

$$\int_{-\infty}^{\infty}\mathfrak{S}_x(z)\overline{\mathfrak{S}_y(z)}\,dz = g(x-y)-g(x)-g(-y)+g(0).$$

We note that, as in [14, (1.5), (1.6)], the function $\mathfrak{S}_x(z)$ can also be represented without reference to Γ, just as g can.

7.8. **Heuristics for formula** (1.12)**.** Formally, we derive (1.12) by following the argument in Section 6 for $\mathcal{H}(A_\infty)$ under the assumption of RH.

Let $K(w,z)$ denote the reproducing kernel of $\mathcal{B}$, namely, $\langle F, K(w,\cdot)\rangle_{\mathcal{B}} = F(w)$. Then

$$K(w,z) = \frac{E(z)\overline{E(w)} - E^\sharp(z)\overline{E^\sharp(w)}}{2\pi i(\bar{w}-z)}, \quad E(z) = \xi(1/2-iz) + \xi'(1/2-iz),$$

since $\mathcal{B}$ is the de Branges space generated by E ([14, Theorem 1.1]). The function $K(w,\cdot)$ is an eigenfunction of M^* corresponding to the eigenvalue $\bar{w}$. Indeed, for every $F \in \mathfrak{D}(M)$, we have

$$\langle MF, K(w,\cdot)\rangle_{\mathcal{B}} = wF(w) = w\langle F, K(w,\cdot)\rangle_{\mathcal{B}} = \langle F, \bar{w}K(w,\cdot)\rangle_{\mathcal{B}}.$$

Since $\mathfrak{D}(M)$ is dense in $\mathcal{B}$ ([14, Section 6]), it follows that $K(w,\cdot) \in \mathfrak{D}(M^*)$ and $M^*K(w,\cdot) = \bar{w}K(w,\cdot)$. In particular, $K(\mp i,\cdot)$ belong to the corresponding deficiency spaces $\mathrm{Ker}(M^* \mp i)$. Furthermore,

$$\|K(\mp i,\cdot)\|_{\mathcal{B}}^2 = \langle K(\mp i,\cdot), K(\mp i,\cdot)\rangle_{\mathcal{B}} = K(\mp i, \mp i) = \xi(3/2)\xi'(3/2)/\pi.$$

Set $W_\theta(t) := K(-i,t) + e^{i\theta}K(i,t)$. For the boundary form $W(F,G) = \langle M^*F, G\rangle_{\mathcal{B}} - \langle F, M^*G\rangle_{\mathcal{B}}$, we obtain

$$\begin{aligned} W(K(\bar{z},\cdot), W_\theta) &= (z+i)\langle K(\bar{z},\cdot), K(-i,\cdot)\rangle_{\mathcal{B}} + e^{-i\theta}(z-i)\langle K(\bar{z},\cdot), K(i,\cdot)\rangle_{\mathcal{B}} \\ &= (z+i)K(\bar{z},-i) + e^{-i\theta}(z-i)K(\bar{z},i) = -\frac{e^{-i\theta/2}}{\pi i}\big(C_\theta E(z) - \overline{C_\theta}E^\sharp(z)\big), \end{aligned}$$

where $C_\theta = \xi(3/2)\cos(\theta/2) + i\xi'(3/2)\sin(\theta/2)$. In particular, when $\theta = \pi$, the right-hand side becomes $(2i/\pi)\,\xi'(3/2)\xi(1/2-iz)$.

Let f_z be the function satisfying $\widehat{f_z}(t) = K(\bar{z},t)/E(t)$. Then f_z belongs to a certain subspace $V(0)$ of $L^2(0,\infty)$ ([14, Lemma 5.1]), and

$$K(\bar{z}, \mp i) = \langle K(\bar{z},\cdot), K(\mp i,\cdot)\rangle_{\mathcal{B}} = \overline{\langle K(\mp i,\cdot), K(\bar{z},\cdot)\rangle_{\mathcal{B}}} = \overline{K(\mp i, \bar{z})} = E^\sharp(z)\overline{\widehat{f_{\pm i}}(\bar{z})}.$$

On the other hand,

$$\langle K(\bar{z},\cdot), K(\mp i,\cdot)\rangle_{\mathcal{B}} = \langle K(\bar{z},\cdot)/E, K(\mp i,\cdot)/E\rangle_{L^2} = 2\pi\langle f_z, f_{\pm i}\rangle_{L^2} = \pi\langle f_z, f_{\pm i}\rangle_{A_\infty},$$

where the last equality follows from [14, Theorem 5.5]. Therefore,

$$\begin{aligned} -\frac{e^{-i\theta/2}}{\pi i}(C_\theta E(z) - \overline{C_\theta}E^\sharp(z)) &= W(K(\bar{z},\cdot), W_\theta) \\ &= \pi(z+i)\langle f_z, f_{+i}\rangle_{A_\infty} + \pi e^{-i\theta}(z-i)\langle f_z, f_{-i}\rangle_{A_\infty} \\ &= \pi E^\sharp(z)(z+i)\overline{\widehat{f_{+i}}(\bar{z})} + \pi E^\sharp(z)e^{-i\theta}(z-i)\overline{\widehat{f_{-i}}(\bar{z})}. \end{aligned}$$

Hence,

$$(z-i)\widehat{f_{+i}}(z) + e^{i\theta}(z+i)\widehat{f_{-i}}(z) = \frac{ie^{i\theta/2}}{\pi^2}(C_\theta - \overline{C_\theta}\frac{E^\sharp(z)}{E(z)}).$$

In particular, when $\theta = \pi$,

$$(z-i)\widehat{f_{+i}}(z) - (z+i)\widehat{f_{-i}}(z) = \frac{2\xi'(3/2)}{\pi^2 i}\frac{\xi(1/2-iz)}{E(z)}.$$

Since $\|f_{+i}\|_{A_\infty} = \|f_{-i}\|_{A_\infty}$ by definition, this identity strongly suggests (1.12).

Furthermore, if $v \in V(0)$, then

$$\langle v, f_z\rangle_{A_\infty} = 2\langle v, f_z\rangle_{L^2} = \frac{1}{\pi}\langle \widehat{v}, \widehat{f_z}\rangle_{L^2} = \frac{1}{\pi}\langle E\widehat{v}, K(\bar{z},\cdot)\rangle_{\mathcal{B}} = \frac{1}{\pi}E(\bar{z})\widehat{v}(\bar{z}).$$

Hence, the analogue in $\mathcal{H}(A_\infty)$ of the vector v_z in Section 6 is given by

$$v_z := \frac{\pi f_z}{E(\bar{z})}.$$

8. From distribution kernels to continuous kernels

8.1. **Passing from A_a to G_a.** We have developed the theory of Q_W^a and A_a using the screw function associated with $\zeta(s)$. However, in the theory of screw functions, as presented for example in [8, Section 5], it is more natural to consider the integral operator $\widetilde{G}$ defined by

$$(\widetilde{G}u)(x) := \int_{-\infty}^{\infty} (g(x-y) - g(x) - g(-y) + g(0))u(y)\,dy \tag{8.1}$$

rather than the integral operator G in (1.4). As shown in [13, Theorem 1.3], if $Q_{\widetilde{G}}(v) := \langle \widetilde{G}v, v\rangle_{L^2}$ denotes the quadratic form associated with $\widetilde{G}$, then the condition $Q_{\widetilde{G}}(u) \geq 0$ for all $u \in C_c^\infty(\mathbb{R}) \cap L_0^2(-a,a)$ is equivalent to RH. This follows from the relation $Q_W(v) = Q_{\widetilde{G}}(Dv)$, $v \in C_c^\infty(\mathbb{R})$ [13, Proposition 3.1]. Note that if $u = Dv$ with $v \in C_c^\infty(\mathbb{R})$, then $u \in L_0^2(-a,a)$. If $Q_G(v) := \langle Gv, v\rangle_{L^2}$ denotes the quadratic form associated with the integral operator G in (1.4), then $Q_{\widetilde{G}}(u) = Q_G(u)$ for every $u \in L_0^2(-a,a)$, and hence $Q_W(v) = Q_{\widetilde{G}}(Dv) = Q_G(Dv)$. Motivated by this relation, we chose to study Q_W through the quadratic form Q_G, whose kernel is considerably simpler.

To represent Q_W directly on the finite interval $(-a,a)$, one must use the operator A_a as in (1.1). However, if one is interested only in the positivity of Q_W, there is no essential difference in working with G_a, since $Q_W(v) = Q_G(Dv)$ and the restriction of Q_G to $(-a,a)$ is represented by the operator G_a through $(Q_G|_{L^2(-a,a)})(u) = \langle G_a u, u\rangle_{L^2}$. Moreover, G_a is an integral operator with a continuous kernel and is therefore much easier to handle than the unbounded operator A_a. As we shall see below, the Hilbert space obtained by completing with respect to $\langle A_a u, u\rangle_{L^2}$ and the Hilbert space obtained by completing with respect to $\langle G_a u, u\rangle_{L^2}$ are related through the identity $Q_W(v) = Q_G(Dv)$, and Theorem 1.5 can be interpreted within the framework of G_a.

Although A_a and G_a are very different as operators, the former being an unbounded operator with discrete spectrum accumulating at $+\infty$ and the latter being a compact operator with discrete spectrum accumulating at 0, the situation is analogous from the viewpoint of positivity of quadratic forms, since in both cases the relevant quantity is the bottom of the spectrum. (There is, of course, the distinction that the infimum may or may not be attained.) Furthermore, as we shall see below, this difference does not affect the spectrum of the corresponding self-adjoint extensions of the differential operator.

To discuss these matters, we first review the relation between the norms arising through the differential operator D.

8.2. **The Inverse Neumann Laplacian.** The operator $D : H_0^1(-a,a) \to L_0^2(-a,a)$ is bijective. Therefore, if $u = Dv$ with $v \in H_0^1(-a,a)$, then $v = D^{-1}u$, and hence

$$\|v\|_{L^2}^2 = \langle D^{-1}u, D^{-1}u\rangle_{L^2} = \langle (DD^*)^{-1}u, u\rangle_{L^2}.$$

Note that $-\Delta_N = DD^*$ coincides with the Neumann Laplacian, which is the operator acting as $-\Delta_N = -d^2/dx^2$ with domain $\mathfrak{D}(-\Delta_N) = \{u \in H^2(-a,a) \mid u'(a) = u'(-a) = 0\}$, and its range is $L_0^2(-a,a)$. Its inverse $(-\Delta_N)^{-1} : L_0^2(-a,a) \to L^2(-a,a)$ is a compact, self-adjoint, and positive integral operator with kernel

$$N(x,y) = \frac{x^2+y^2}{4a} - \frac{|x-y|}{2} + \frac{a}{6}.$$

On the other hand, if $u = Dv$ with $v \in H_0^1(-a,a)$, then $v(x) = \int_{-a}^x u(t)dt$, and therefore

$$\begin{aligned}\|v\|_{L^2}^2 &= \int_{-a}^a |v(x)|^2 dx = \int_{-a}^a \left(\int_{-a}^a u(t)\mathbf{1}_{t\leq x} dt\right)\left(\int_{-a}^a \overline{u(s)}\mathbf{1}_{s\leq x} ds\right) dt \\ &= \int_{-a}^a\int_{-a}^a \left(\int_{\max(t,s)}^a dx\right) u(t)\overline{u(s)}dtds = \int_{-a}^a\int_{-a}^a (a - \max(t,s))u(t)\overline{u(s)}dtds.\end{aligned}$$

Accordingly, defining

$$(\widetilde{K}_a u)(x) := \int_{-a}^{a} (a - \max(x,y))u(y)dy,$$

we obtain

$$\|v\|_{L^2}^2 = \langle \widetilde{K}_a u, u\rangle_{L^2}.$$

The operator $\widetilde{K}_a$ is compact, self-adjoint, and positive. Moreover, the identity

$$\|v\|_{L^2}^2 = \langle \widetilde{K}_a u, u\rangle_{L^2} = \langle (-\Delta_N)^{-1} u, u\rangle_{L^2} \tag{8.2}$$

holds as an equality of quadratic forms on $L_0^2(-a,a)$. Since the operator-theoretic properties of $(-\Delta_N)^{-1}$ are more transparent, we shall henceforth write

$$K_a := (-\Delta_N)^{-1}, \quad \text{with integral kernel } N(x,y) = \frac{x^2+y^2}{4a} - \frac{|x-y|}{2} + \frac{a}{6},$$

and work with K_a rather than $\widetilde{K}_a$.

8.3. **The Hilbert Space $\mathcal{H}(S_a)$.** Let $\lambda < \lambda_a$. Motivated by (8.2) and by the operator $T_a = A_a - \lambda I$ considered earlier, we define

$$S_a := G_a - \lambda(-\Delta_N)^{-1}$$

on $C_c^\infty(-a,a) \cap L_0^2(-a,a)$. Since both G_a and $(-\Delta_N)^{-1}$ are self-adjoint with respect to the L^2 inner product, so is S_a, and

$$\|u\|_{S_a}^2 = \langle S_a u, u\rangle_{L^2} = \langle T_a v, v\rangle_{L^2} = \|v\|_{T_a}^2, \quad u = Dv \in C_c^\infty(-a,a) \cap L_0^2(-a,a).$$

Furthermore,

$$\mathcal{H}(S_a) := \Big[\text{the completion of } C_c^\infty(-a,a) \cap L_0^2(-a,a) \text{ with respect to } \|u\|_{S_a}\Big]$$

is isometrically isomorphic to $\mathcal{H}(T_a)$, since $D : C_c^\infty(-a,a) \to C_c^\infty(-a,a) \cap L_0^2(-a,a)$ is bijective. The operator D therefore extends to an isometric isomorphism

$$\bar{D} : \mathcal{H}(T_a) \overset{\sim}{\to} \mathcal{H}(S_a).$$

Note that $\mathcal{H}(T_a) \hookrightarrow L^2(-a,a)$, whereas $\mathcal{H}(S_a) \not\subset L^2(-a,a)$. It should also be noted that $\mathbf{1}_{[-a,a]} \in \mathcal{H}(T_a) \setminus \mathfrak{D}(\mathscr{D}_a)$, while $\bar{D}(\mathbf{1}_{[-a,a]}) \neq 0$. Through the isomorphism $\bar{D}$, operators on $\mathcal{H}(T_a)$ can be transferred to operators on $\mathcal{H}(S_a)$. Accordingly, we define an operator $\widetilde{\mathscr{D}}_a$ on $\mathcal{H}(S_a)$ by

$$\widetilde{\mathscr{D}}_a := D\mathscr{D}_a D^{-1}$$

with domain $C_c^\infty(-a,a) \cap L_0^2(-a,a)$. For $u \in C_c^\infty(-a,a) \cap L_0^2(-a,a)$,

$$(D^{-1}u)(x) = -i\int_{-a}^{x} u(t)\,dt, \quad D\mathscr{D}_a(D^{-1}u)(x) = iu'(x),$$

and hence $\widetilde{\mathscr{D}}_a$ may be regarded as the operator with domain $C_c^\infty(-a,a) \cap L_0^2(-a,a)$ acting by $\widetilde{\mathscr{D}}_a = i\,d/dx$. By construction, $\widetilde{\mathscr{D}}_a$ on $\mathcal{H}(S_a)$ is unitarily equivalent to $\mathscr{D}_a$ on $\mathcal{H}(T_a)$. Therefore, $\widetilde{\mathscr{D}}_a$ has deficiency indices $(1,1)$, and its self-adjoint extensions have the same spectra as those of $\mathscr{D}_a$.

We extend the operator S_a to $\mathcal{H}(S_a)$ via the unitary isomorphism $\bar{D}$, namely,

$$S_a = \bar{D}\,T_a(\bar{D})^{-1},$$

where S_a denotes the self-adjoint operator associated with the closed symmetric quadratic form induced by $G_a - \lambda(-\Delta_N)^{-1}$. Then, $\mathscr{D}_a^* v_z = zv_z$ is equivalent to $\widetilde{\mathscr{D}}_a^* u_z = zu_z$ for $u_z := \bar{D}v_z$, since $\mathscr{D}_a^* = \bar{D}^{-1}\widetilde{\mathscr{D}}_a^*\bar{D}$. We define the corresponding boundary form

$$\widetilde{W}(u,v) := \langle \widetilde{\mathscr{D}}_a^* u, v\rangle_{S_a} - \langle u, \widetilde{\mathscr{D}}_a^* v\rangle_{S_a}$$

in analogy with (6.1). Setting $u_\pm := \bar{D}v_\pm$ and $\widetilde{w}_\theta := u_+ + e^{i\theta}u_-$, we obtain

$$\widetilde{W}(u_z, \widetilde{w}_\theta) = (z+i)\langle u_z, u_+\rangle_{S_a} + e^{-i\theta}(z-i)\langle u_z, u_-\rangle_{S_a}. \tag{8.3}$$

Since $\bar{D}$ is unitary, it follows from (6.3) that

$$\langle u_z, u_\pm\rangle_{S_a} = \langle \bar{D}v_z, \bar{D}v_\pm\rangle_{S_a} = \langle v_z, v_\pm\rangle_{T_a} = \overline{\widehat{v_\pm}(\bar{z})}.$$

Hence, by (1.11), $W(a,\theta;z) = \overline{W(v_{\bar{z}}, w_\theta)} = \overline{\widetilde{W}(u_{\bar{z}}, \widetilde{w}_\theta)}$. Thus the boundary forms determining the eigenvalues coincide for $\widetilde{\mathscr{D}}_a^*$ and $\mathscr{D}_a^*$. For $e_z(x) := \exp(-izx)$, the equation $T_a v_z = e_z$ becomes $S_a u_z = \bar{D}e_z$.

Let $k(x,y) = g(x-y) - \lambda N(x,y)$. Then the equations $T_a v_\pm = C_\pm e_{\pm i}$ may be written as

$$\int_{-a}^{a} (-k_{xx}(x,y))v_\pm(y)\,dy = C_\pm e^{\pm x}, \qquad x \in (-a,a). \tag{8.4}$$

The equations (8.4) are formally equivalent to

$$\int_{-a}^{a} k(x,y)(-v_\pm(y))\,dy = C_\pm e^{\pm x} + A_\pm x + B_\pm, \qquad x \in (-a,a) \tag{8.5}$$

in the sense that differentiating both sides twice with respect to x yields (8.4), where

$$A_\pm = \int_{-a}^{a} k_x(0,y)(-v_\pm(y))\,dy \mp C_\pm, \qquad B_\pm = \int_{-a}^{a} k(0,y)(-v_\pm(y))\,dy - C_\pm.$$

Here we avoid expressing these equations in terms of the operators T_a or S_a, since the above argument ignores domain issues. It should also be noted that the equation (8.5) is different from $S_a u_\pm = C_\pm \bar{D}e_{\pm i}$.

Since the kernel in (8.5) is continuous, it is an ordinary Fredholm integral equation of the first kind. This observation suggests a numerical approach. One may compute $v_\pm(a,x)$ by solving this Fredholm equation numerically and then test experimentally whether (1.12) holds. Our numerical experiments provide evidence supporting (1.12) for the choice $\theta = \pi$.

8.4. **Relation to Bombieri's Problem 1.** Using $\widetilde{G}$ in (8.1) and P_a in (2.1), define

$$\widetilde{G}_a := P_a\widetilde{G}P_a : L_0^2(-a,a) \to L_0^2(-a,a).$$

Recall that $G_a = P_aGP_a$ in (1.5), and $B_a = D^*G_aD$ in (1.6). Then

$$\langle \widetilde{G}_aDv, Dv\rangle_{L^2} = \langle G_aDv, Dv\rangle_{L^2} = \langle B_av, v\rangle_{L^2} = Q_W(v), \quad v \in C_c^\infty(-a,a), \tag{8.6}$$

see [13, Proposition 3.1]. For finite a, the norm on $H_0^1(-a,a)$ defined by $\|v\|_{L^2} + \|v'\|_{L^2}$ is equivalent to that defined by $\|v'\|_{L^2}$. Adopting the latter, the linear map $H_0^1(-a,a) \to L_0^2(-a,a)$ given by $v \mapsto Dv$ becomes an isometric isomorphism. Hence the Rayleigh quotient $Q_W(v)/\|v\|_{H^1}^2$ on $H_0^1(-a,a)$ appearing in [1, Problem 1] (and [2, Problem B]) can be rewritten as the Rayleigh quotient on $L_0^2(-a,a)$

$$\frac{Q_W(v)}{\|v\|_{H^1}^2} = \frac{Q_G(u)}{\|u\|_{L^2}^2}, \qquad u = Dv,$$

by [13, Proposition 3.1]. In other words, [1, Problem 1] (and [2, Problem B]) is nothing but the eigenvalue problem for $G_a = P_aGP_a = P_a\widetilde{G}P_a$ on $L_0^2(-a,a)$. [1, Theorem 4] asserts that if this infimum is negative, then it is attained in $L^2(-a,a)$. From the viewpoint of the eigenvalue problem for G_a, this is immediate.

Since Q_W is bounded with respect to the H^1-norm, the Riesz representation theorem implies that it is represented by a bounded operator. That operator is precisely $\widetilde{G}_a = G_a$ (they coincide on $L_0^2(-a,a)$). Problem 1 in [1] corresponds to this formulation.

8.5. **Reformulation as a generalized eigenvalue problem.** By (8.2),

$$\|v\|_{L^2}^2 = \langle K_a u, u\rangle_{L^2}$$

with $u = Dv$ and $K_a = (-\Delta_N)^{-1}$. Combining this with (8.6), the Rayleigh quotient in (1.7) can be rewritten as

$$\frac{Q_W^a(v)}{\|v\|_{L^2}^2} = \frac{\langle G_a u, u\rangle_{L^2}}{\langle K_a u, u\rangle_{L^2}} \tag{8.7}$$

for $v \in H_0^1(-a,a)$ and $u = Dv \in L_0^2(-a,a)$. The right-hand side of (8.7) leads to the generalized eigenvalue problem

$$G_a u = \lambda K_a u, \qquad u \in \mathcal{H}(S_a).$$

The spectrum of this generalized eigenvalue problem coincides with that of A_a. Hence it is discrete, bounded from below, and accumulates only at $+\infty$. (The case $\lambda = 0$ is exceptional: then the contribution of K_a disappears, and one is simply looking at the kernel (the 0-eigenspace) of G_a.)

Although A_a is unbounded, both G_a and K_a are compact operators with continuous integral kernels. In the generalized eigenvalue problem, one must analyze not only the spectra of G_a and K_a individually but also the interaction between them. Thus it cannot be said a priori that the spectral analysis of A_a becomes simpler. Nevertheless, the availability of compact-operator techniques may provide certain advantages.

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